Q. Create the standard calculator application in android:

Adroidmanifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">

  <application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.Calculator">
    <activity
      android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
  </application>

</manifest>
```

Colors

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <color name="black">#FF000000</color>
  <color name="white">#FFFFFFFF</color>
  <color name="teal_primary">#009688</color>
  <color name="teal_dark">#00796B</color>
  <color name="button_gray">#E0E0E0</color>
</resources>
```

Themes

```
<resources xmlns:tools="http://schemas.android.com/tools">
  <style name="Base.Theme.Calculator" parent="Theme.AppCompat.Light.DarkActionBar">
    <item name="colorPrimary">@color/teal_primary</item>
    <item name="colorPrimaryDark">@color/teal_dark</item>
    <item name="colorAccent">@color/teal_primary</item>
```

```
        <item name="titleTextColor">@color/white</item>
    </style>

    <style name="Theme.Calculator" parent="Base.Theme.Calculator" />

    <style name="CalcBtn">
        <item name="android:layout_width">0dp</item>
        <item name="android:layout_height">0dp</item>
        <item name="android:layout_columnWeight">1</item>
        <item name="android:layout_rowWeight">1</item>
        <item name="android:textSize">20sp</item>
        <item name="android:textStyle">bold</item>
        <item name="android:layout_margin">6dp</item>
        <item name="android:background">@color/button_gray</item>
        <item name="android:textColor">@color/black</item>
    </style>
</resources>
```

Activity main

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp"
    android:background="#FFFFFF">

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="250dp"
        android:background="#EEEEEE"
        android:gravity="bottom|end"
        android:padding="50dp"
        android:text="0"
        android:textColor="#000000"
        android:textSize="48sp" />

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:columnCount="4"
        android:rowCount="5"
```

```
        android:layout_marginTop="10dp"
        android:alignmentMode="alignMargins">

        <Button android:id="@+id/btnDel" android:text="DEL" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnClear" android:text="C" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnMod" android:text="%" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnAdd" android:text="+" style="@style/CalcBtn"/>

        <Button android:text="1" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="2" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="3" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnDiv" android:text="/" style="@style/CalcBtn"/>

        <Button android:text="4" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="5" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="6" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnSub" android:text="-" style="@style/CalcBtn"/>

        <Button android:text="7" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="8" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="9" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:id="@+id/btnMul" android:text="X" style="@style/CalcBtn"/>

        <Button android:text="." android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button android:text="0" android:onClick="onDigitClick" style="@style/CalcBtn"/>
        <Button
            android:id="@+id/btnEqual"
            android:text="="
            style="@style/CalcBtn"
            android:layout_columnSpan="2"
            android:layout_gravity="fill_horizontal"/>

    </GridLayout>
</LinearLayout>
```

Main activity

```
package com.example.calculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
```

```java
public class MainActivity extends AppCompatActivity {

    private TextView tvResult;
    private String currentInput = "";
    private double firstValue = 0.0;
    private String operator = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvResult = findViewById(R.id.tvResult);

        // --- Setup Operator Buttons ---
        setOperatorListener(R.id.btnAdd, "+");
        setOperatorListener(R.id.btnSub, "-");
        setOperatorListener(R.id.btnMul, "*");
        setOperatorListener(R.id.btnDiv, "/");
        setOperatorListener(R.id.btnMod, "%");

        // --- Setup Functional Buttons ---
        // Clear Button (C)
        findViewById(R.id.btnClear).setOnClickListener(v -> {
            currentInput = "";
            firstValue = 0.0;
            operator = "";
            tvResult.setText("0");
        });

        // Delete Button (DEL)
        findViewById(R.id.btnDel).setOnClickListener(v -> {
            if (currentInput.length() > 0) {
                currentInput = currentInput.substring(0, currentInput.length() - 1);
                tvResult.setText(currentInput.isEmpty() ? "0" : currentInput);
            }
        });

        // Equals Button (=)
        findViewById(R.id.btnEqual).setOnClickListener(v -> calculateResult());
    }

    // Helper method to handle Number clicks (linked via XML 'onClick')
    public void onDigitClick(View view) {
```

```
        Button btn = (Button) view;
        currentInput += btn.getText().toString();
        tvResult.setText(currentInput);
    }

    // Helper to set listeners for math operators
    private void setOperatorListener(int btnId, String op) {
        findViewById(btnId).setOnClickListener(v -> {
            if (!currentInput.isEmpty()) {
                firstValue = Double.parseDouble(currentInput);
                operator = op;
                currentInput = ""; // Clear input for the second number
            }
        });
    }

    private void calculateResult() {
        if (!currentInput.isEmpty() && !operator.isEmpty()) {
            double secondValue = Double.parseDouble(currentInput);
            double result = 0.0;

            switch (operator) {
                case "+": result = firstValue + secondValue; break;
                case "-": result = firstValue - secondValue; break;
                case "*": result = firstValue * secondValue; break;
                case "/": result = firstValue / secondValue; break;
                case "%": result = firstValue % secondValue; break;
            }

            // Update display and reset for next operation
            tvResult.setText(String.valueOf(result));
            currentInput = String.valueOf(result);
            operator = "";
        }
    }
}
```

Q. Create a login screen with EditText fields for username and password, and a Button to submit. Include validation for empty fields.

Android manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  package="com.example.uielements">
  <application
     android:allowBackup="true"
     android:icon="@mipmap/ic_launcher"
     android:label="@string/app_name"
     android:roundIcon="@mipmap/ic_launcher_round"
     android:supportsRtl="true"
     android:theme="@style/Theme.UIElements">
     <activity android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
           <action android:name="android.intent.action.MAIN" />
           <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
     </activity>
  </application>
</manifest>
```

Main activity

```
package com.example.uielements;


import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

  @Override

  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
     // Setting up Toolbar as AppBar
     Toolbar toolbar = findViewById(R.id.toolbar);
```

```
        setSupportActionBar(toolbar);
        // Reference UI elements
        EditText et_user_name = findViewById(R.id.et_user_name);
        EditText et_password = findViewById(R.id.et_password);
        Button btnReset = findViewById(R.id.btn_reset);
        Button btnSubmit = findViewById(R.id.btn_submit);
        // Reset Button functionality
        btnReset.setOnClickListener(v -> {
            et_user_name.setText("");
            et_password.setText("");
        });
        // Submit Button functionality
        btnSubmit.setOnClickListener(v -> {
            // 1. Get the text and trim spaces
            String uname = et_user_name.getText().toString().trim();
            String pwd = et_password.getText().toString().trim();

            // 2. Validation: Check if Username is empty
            if (uname.isEmpty()) {
                et_user_name.setError("Username is required");
                et_user_name.requestFocus();
                return; // Stop code here
            }

            // 3. Validation: Check if Password is empty
            if (pwd.isEmpty()) {
                et_password.setError("Password is required");
                et_password.requestFocus();
                return; // Stop code here
            }

            // 4. If validation passes
            Toast.makeText(MainActivity.this, "Welcome " + uname, Toast.LENGTH_LONG).show();
        });
//        // Load Fragment dynamically
//        if (savedInstanceState == null) {
//            FragmentExample fragment = new FragmentExample();
//            FragmentTransaction transaction = getSupportFragmentManager().beginTransaction();
//            transaction.replace(R.id.fragment_container, fragment);
//            transaction.commit();
//        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">
  <!-- AppBar with Toolbar -->
  <com.google.android.material.appbar.AppBarLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:theme="@style/ThemeOverlay.AppCompat.Dark.ActionBar">
    <androidx.appcompat.widget.Toolbar
      android:id="@+id/toolbar"
      android:layout_width="match_parent"
      android:layout_height="?attr/actionBarSize"
      app:title="UI Elements Demo"
      app:titleTextColor="@android:color/white"
      app:popupTheme="@style/ThemeOverlay.AppCompat.Light"/>
  </com.google.android.material.appbar.AppBarLayout>
  <!-- Main Layout for UI Components -->
  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_marginTop="80dp">
    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:textSize="25sp"
      android:text="Login"
      android:layout_gravity="center_horizontal"/>
    <com.google.android.material.textfield.TextInputLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginTop="16dp">

      <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/et_user_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
            android:hint="Username"
            android:inputType="textPersonName" />
        </com.google.android.material.textfield.TextInputLayout>
        <com.google.android.material.textfield.TextInputLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            app:passwordToggleEnabled="true">  <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/et_password"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"
            android:inputType="textPassword" />

        </com.google.android.material.textfield.TextInputLayout>
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:orientation="horizontal"
            android:layout_marginTop="16dp">
            <Button
                android:id="@+id/btn_reset"
                android:text="Reset"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginEnd="16dp"/>
            <Button
                android:id="@+id/btn_submit"
                android:text="Submit"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"/>
        </LinearLayout>
        <!-- Fragment Container -->
        <FrameLayout
            android:id="@+id/fragment_container"
            android:layout_width="match_parent"
            android:layout_height="0dp"
            android:layout_weight="1"
            android:layout_marginTop="16dp"/>
    </LinearLayout>
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

Q. Create an android application to display Alert Dialog on pressing the Back button.

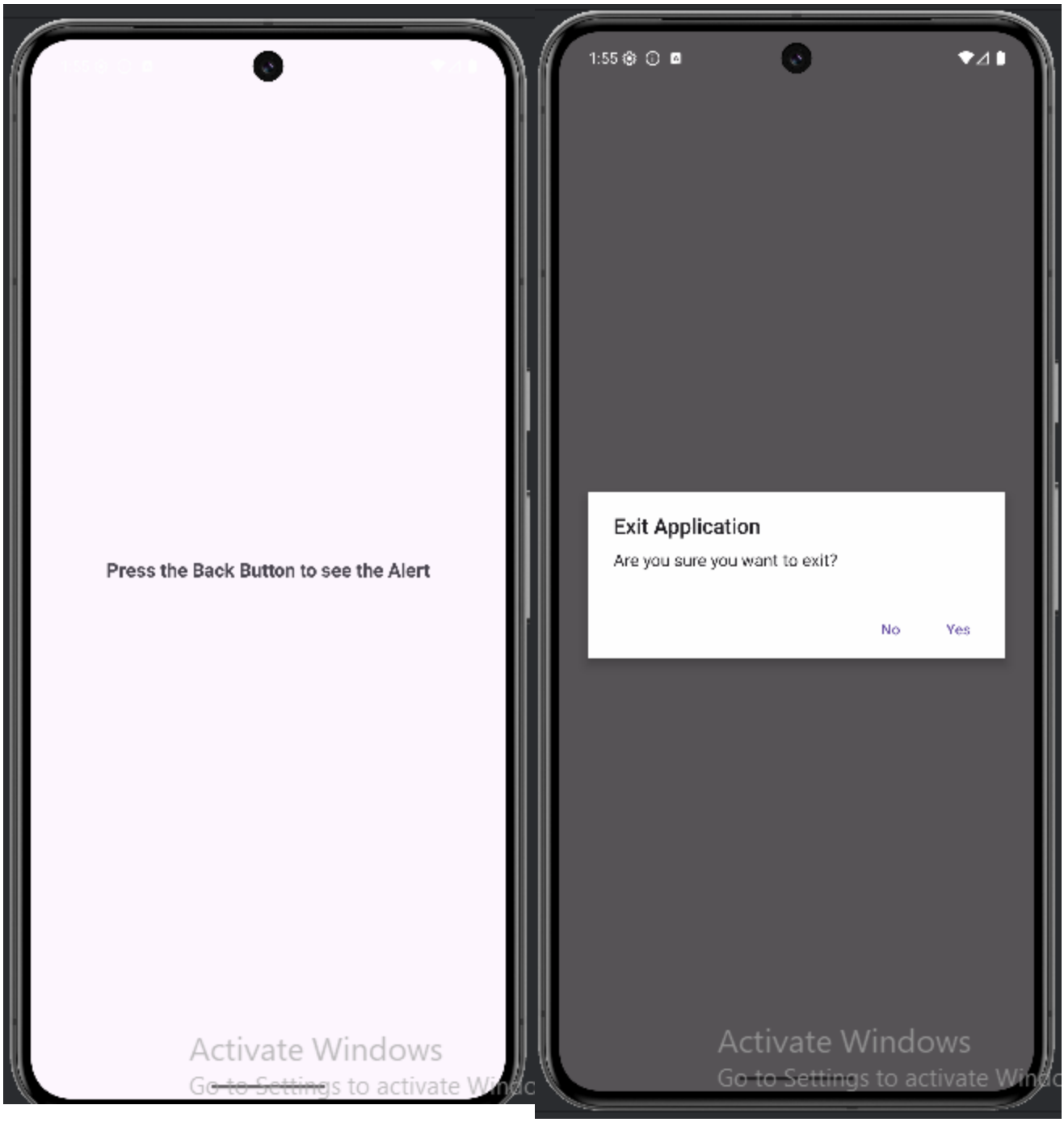
Activity_main.xml

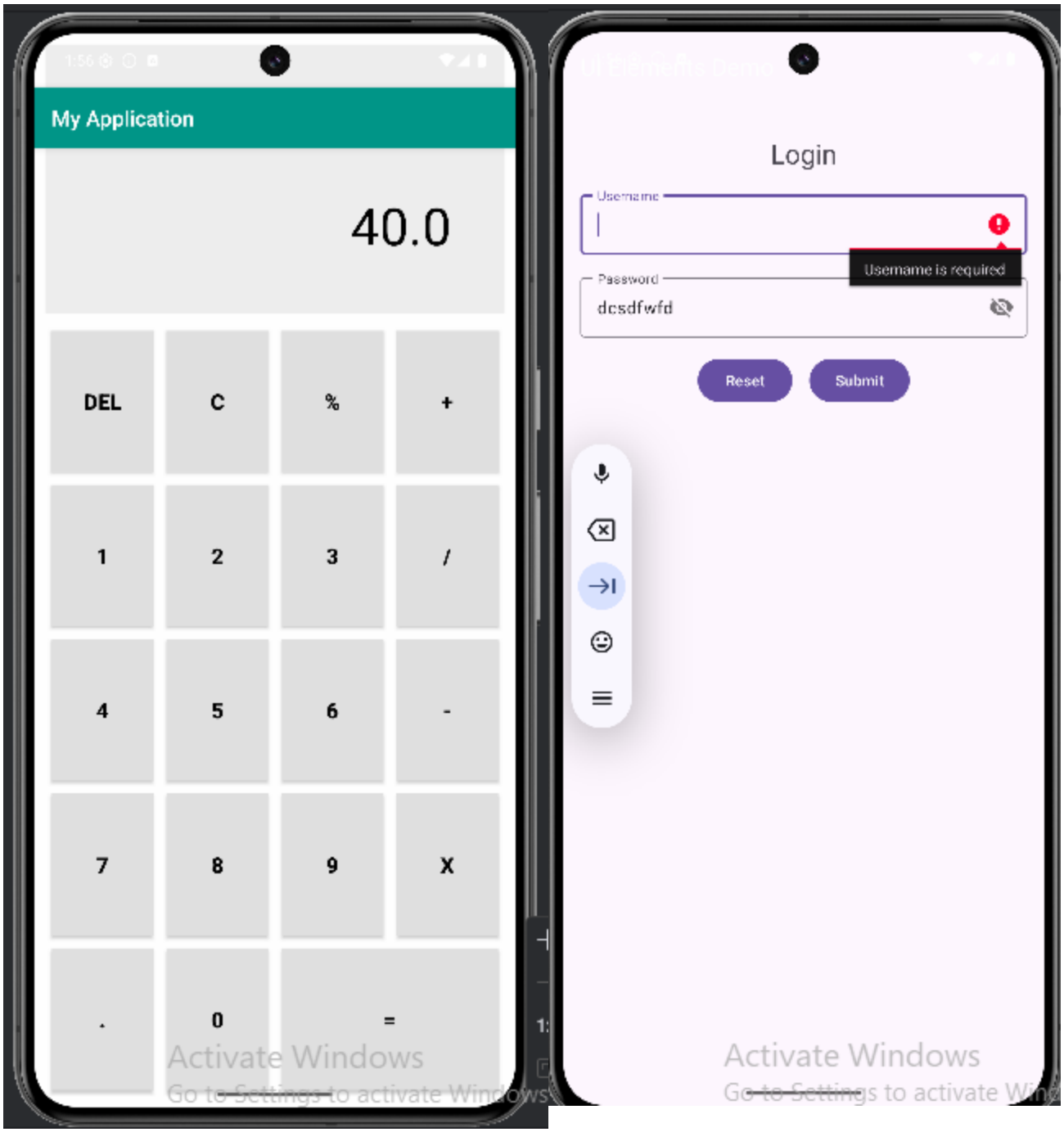
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Press the Back Button to see the Alert"
     android:textSize="18sp"
     android:textStyle="bold" />

</LinearLayout>
```

Main activity

```
package com.example.alertdialogapp;

import android.os.Bundle;
import androidx.activity.OnBackPressedCallback;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);

     // Create a callback to handle the back button press
     OnBackPressedCallback callback = new OnBackPressedCallback(true) {
        @Override
        public void handleOnBackPressed() {
           showExitDialog();
        }
     };
```

```
        // Add the callback to the dispatcher
        getOnBackPressedDispatcher().addCallback(this, callback);
    }

    private void showExitDialog() {
        // Build the Alert Dialog
        new AlertDialog.Builder(this)
                .setTitle("Exit Application")
                .setMessage("Are you sure you want to exit?")
                .setCancelable(false) // Prevent closing by clicking outside
                .setPositiveButton("Yes", (dialog, which) -> {
                    // User clicked Yes, so close the activity
                    finish();
                })
                .setNegativeButton("No", (dialog, which) -> {
                    // User clicked No, so just close the dialog
                    dialog.dismiss();
                })
                .show();
    }
}
```


Press the Back Button to see the Alert
1:55
Exit Application
Are you sure you want to exit?
No
Yes



Q Create an android application to pass the data from one activity to another activity in the same application using intent.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">

  <EditText
     android:id="@+id/etInput"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:hint="Type a message here"
     android:textSize="18sp" />

  <Button
     android:id="@+id/btnSend"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Send to Second Activity"
     android:layout_marginTop="20dp" />

</LinearLayout>
```

Activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">

  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Received Data:"
     android:textSize="16sp" />

  <TextView
     android:id="@+id/tvResult"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Waiting..."
     android:textSize="24sp"
```

```
        android:textStyle="bold"
        android:layout_marginTop="10dp"
        android:textColor="@android:color/holo_blue_dark"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.datapassingapp; // TODO: Check your package name

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // Declare variables
    private EditText etInput;
    private Button btnSend;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 1. Bind Views (These IDs match activity_main.xml)
        etInput = findViewById(R.id.etInput);
        btnSend = findViewById(R.id.btnSend);

        // 2. Set Button Click Listener
        btnSend.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get the string from the EditText
                String message = etInput.getText().toString();

                // Create the Intent object (From Here -> To SecondActivity)
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);

                // Pass the data (Key: "USER_MESSAGE", Value: message)
                intent.putExtra("USER_MESSAGE", message);
```

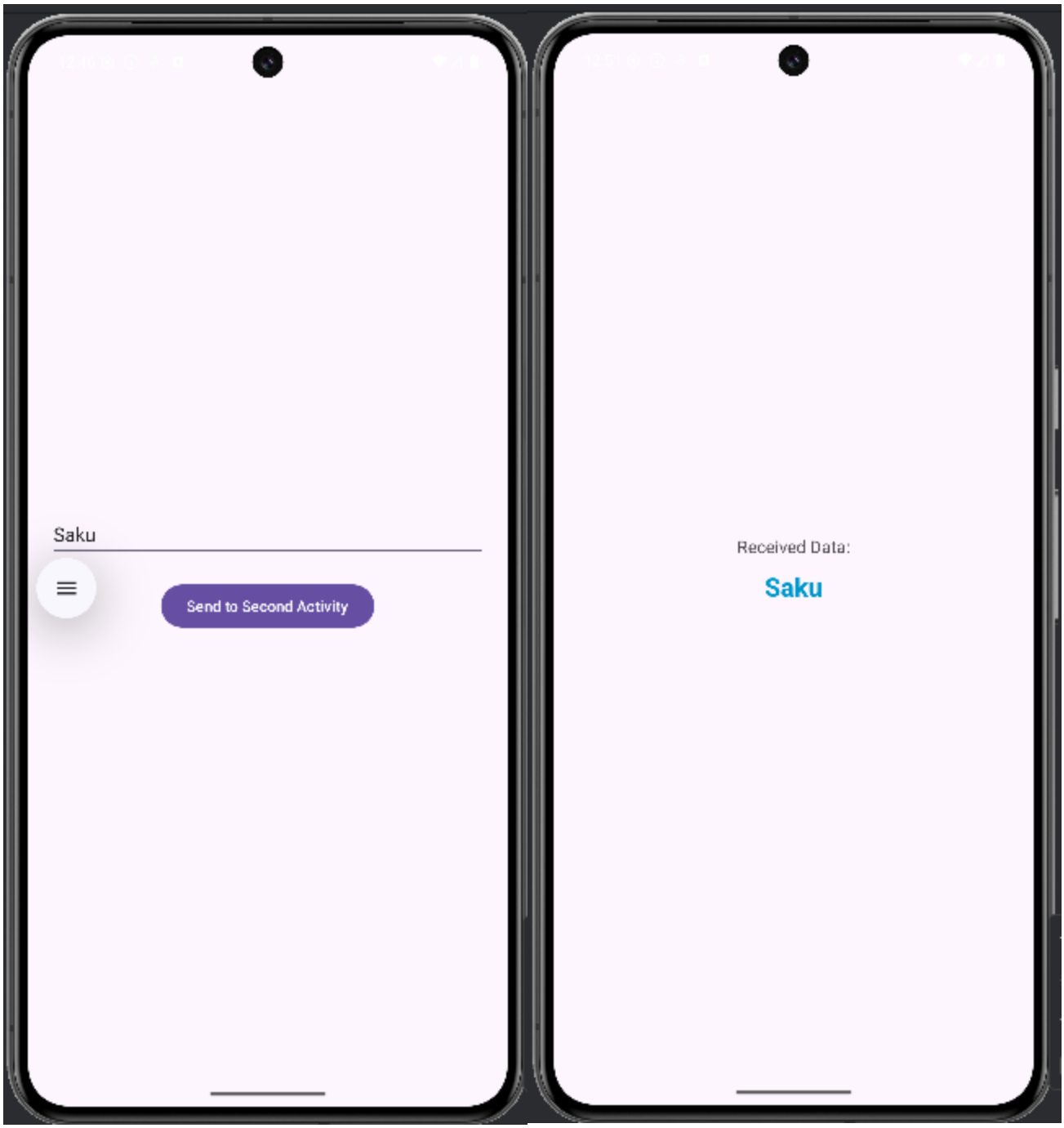
```
            // Start the new activity
            startActivity(intent);
        }
    });
  }
}
```

SecondActivity

```
package com.example.datapassingapp; // TODO: Check your package name

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

  private TextView tvResult;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_second);

    // 1. Bind View (Matches activity_second.xml)
    tvResult = findViewById(R.id.tvResult);

    // 2. Get the Intent that started this activity
    Intent intent = getIntent();

    // 3. Extract the string using the same Key "USER_MESSAGE"
    if (intent != null) {
      String receivedText = intent.getStringExtra("USER_MESSAGE");

      // Set the text to the TextView
      tvResult.setText(receivedText);
    }
  }
}
```

AndroidMAnifest addf this line

```
    <activity android:name=".SecondActivity" />
```



Q. Create a user interface with a TextView, Button, and ImageView. When the button is clicked, the image should be changed dynamically.

Prerequisite: Add Images

Before running this code, you **must** copy two images into your res/drawable folder:

1. Name the first image image_one.jpg (or .png)
2. Name the second image image_two.jpg (or .png)

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="20dp">


    <TextView

        android:id="@+id/tvStatus"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Current Image: One"

        android:textSize="22sp"

        android:textStyle="bold"

        android:layout_marginBottom="20dp"/>


    <ImageView

        android:id="@+id/ivDisplay"

        android:layout_width="250dp"

        android:layout_height="250dp"

        android:src="@drawable/image_one"

        android:scaleType="centerCrop"
```

```
        android:layout_marginBottom="30dp"/>


    <Button

        android:id="@+id/btnChange"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Change Image"

        android:padding="15dp"/>


</LinearLayout>
```

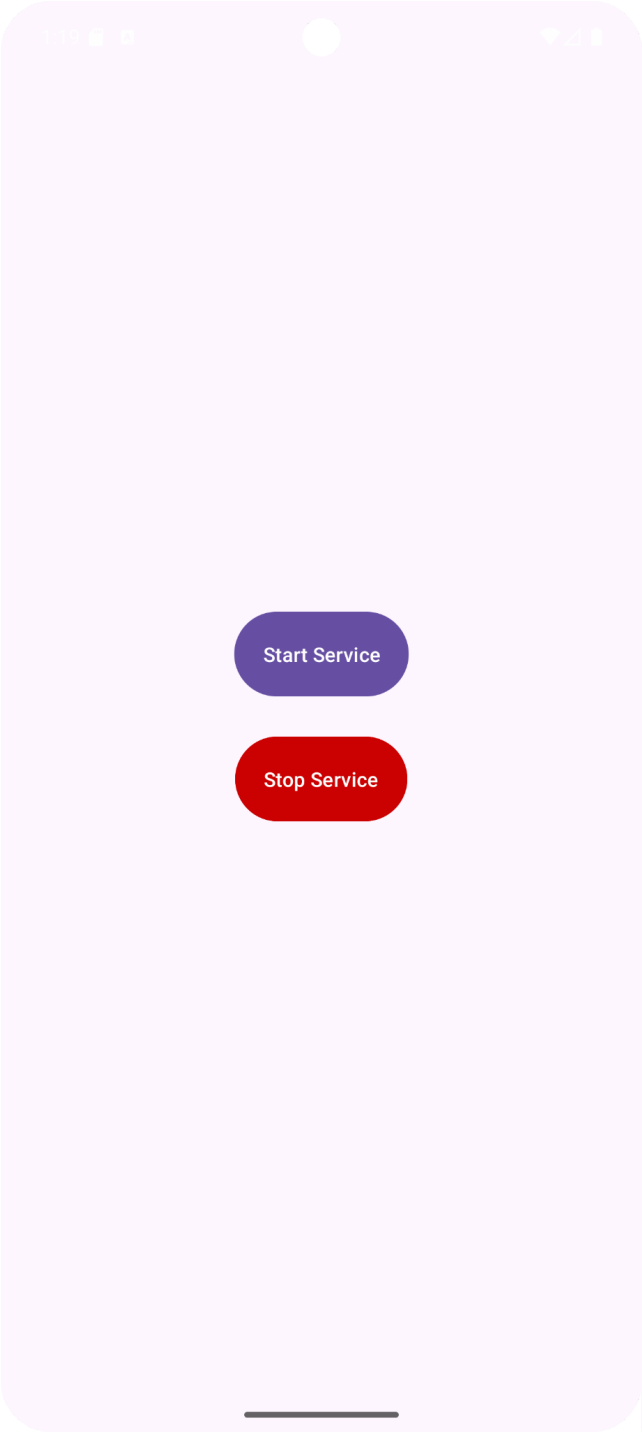
MainActivity.java

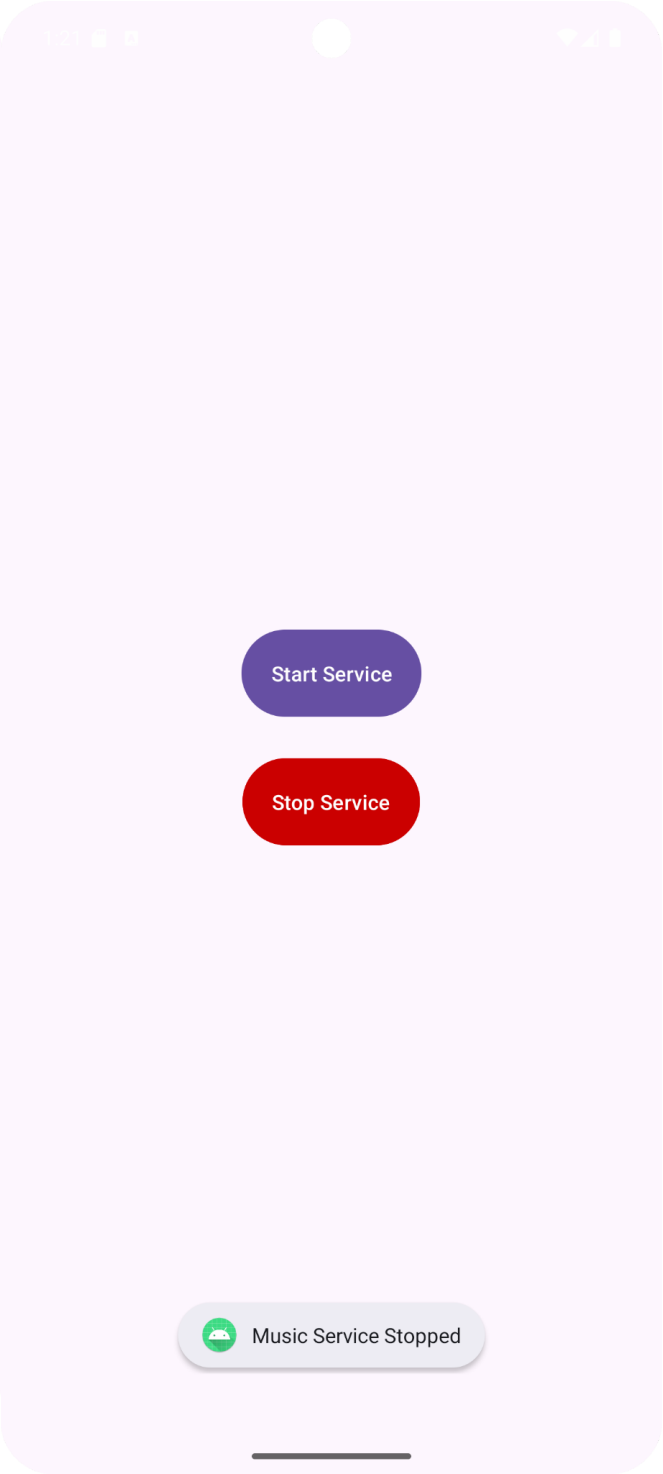
```
package com.example.imagechanger; // TODO: Check your package name


import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.ImageView;

import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    // Declare View variables

    private TextView tvStatus;

    private ImageView ivDisplay;
```

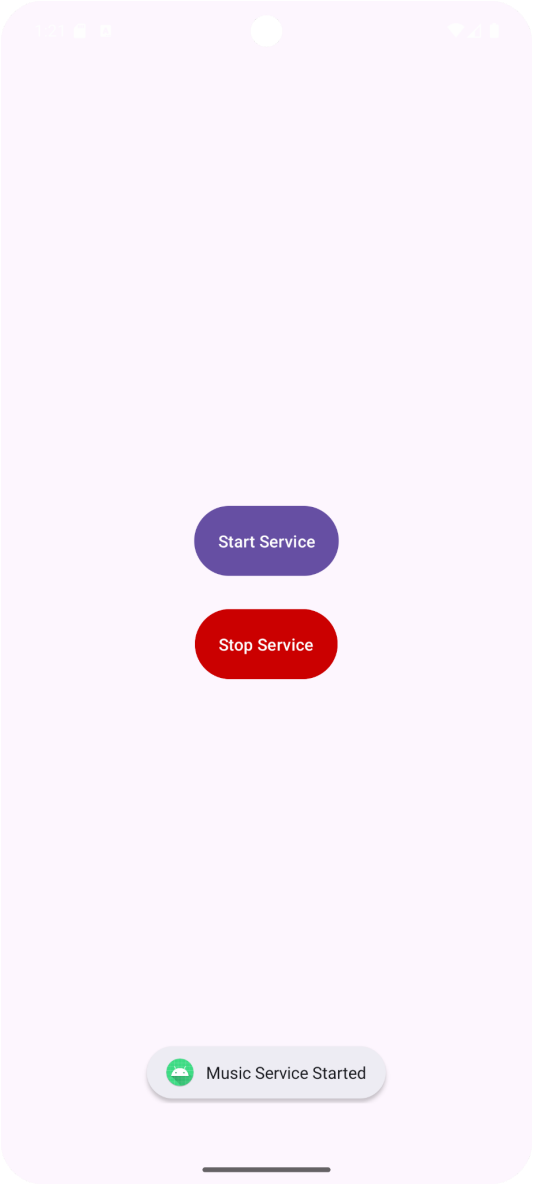
```java
    private Button btnChange;

    // Flag to track which image is currently showing
    private boolean isImageOne = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 1. Initialize Views
        tvStatus = findViewById(R.id.tvStatus);
        ivDisplay = findViewById(R.id.ivDisplay);
        btnChange = findViewById(R.id.btnChange);

        // 2. Set Button Click Listener
        btnChange.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (isImageOne) {
                    // Switch to second image
                    ivDisplay.setImageResource(R.drawable.image_two);
                    tvStatus.setText("Current Image: Two");
                    isImageOne = false; // Update flag
                } else {
                    // Switch back to first image
```

```
                ivDisplay.setImageResource(R.drawable.image_one);

                tvStatus.setText("Current Image: One");

                isImageOne = true; // Update flag

            }

        }

    });

    }

}
```

Q. Create the background service android application to play the ringtone/music.


Start Service
Stop Service


Start Service
Stop Service
Music Service Stopped



res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="20dp">
```

```
    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Service"
        android:padding="20dp" />

    <Button
        android:id="@+id/btnStop"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Stop Service"
        android:padding="20dp"
        android:layout_marginTop="20dp"
        android:backgroundTint="@android:color/holo_red_dark" />

</LinearLayout>
```

MusicService.java

```
package com.example.backgroundmusic; // TODO: Check your package name

import android.app.Service;
import android.content.Intent;
import android.media.MediaPlayer;
import android.os.IBinder;
```

```java
import android.provider.Settings;

import android.widget.Toast;

import androidx.annotation.Nullable;


public class MusicService extends Service {


    private MediaPlayer player;


    @Override

    @Nullable

    public IBinder onBind(Intent intent) {

        // We are not binding to an activity, so return null

        return null;

    }


    @Override

    public int onStartCommand(Intent intent, int flags, int startId) {

        // 1. Create the MediaPlayer logic

        if (player == null) {

            // Use the default system ringtone

            player = MediaPlayer.create(this, Settings.System.DEFAULT_RINGTONE_URI);

            player.setLooping(true); // Loop the music

        }


        // 2. Start the player if it's not already playing

        if (!player.isPlaying()) {
```

```
            player.start();
            Toast.makeText(this, "Music Service Started", Toast.LENGTH_SHORT).show();
        }

        // START_STICKY ensures the service restarts if the system kills it
        return START_STICKY;
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        // 3. Cleanup: Stop and release the player when service is stopped
        if (player != null) {
            if (player.isPlaying()) {
                player.stop();
            }
            player.release();
            player = null;
        }
        Toast.makeText(this, "Music Service Stopped", Toast.LENGTH_SHORT).show();
    }
}
```

MainActivity.java

```
package com.example.backgroundmusic; // TODO: Check your package name

import android.content.Intent;
```

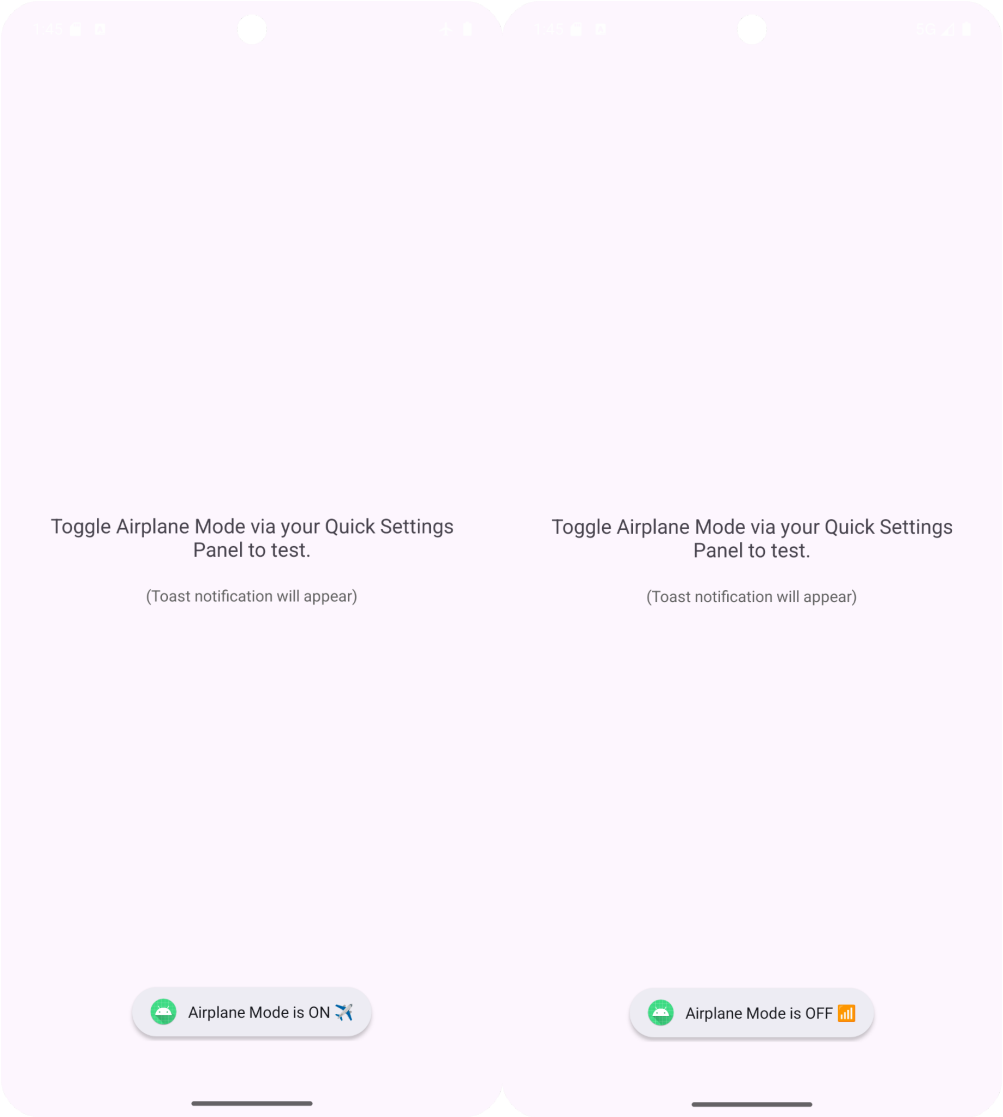
```java
import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    Button btnStart, btnStop;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        btnStart = findViewById(R.id.btnStart);

        btnStop = findViewById(R.id.btnStop);


        // Listener to Start Service

        btnStart.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View v) {

                Intent serviceIntent = new Intent(MainActivity.this, MusicService.class);

                startService(serviceIntent);

            }

        });
```

```
        // Listener to Stop Service
        btnStop.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent serviceIntent = new Intent(MainActivity.this, MusicService.class);
                stopService(serviceIntent);
            }
        });
    }
}
```

Android manifest

```
<service android:name=".MusicService" />
```

Q. Create an android application which automatically notify the user when Aeroplane mode is turned on or off using a broadcast receiver.



MainActivity.java

```
package com.example.airplanemodeapp; // TODO: Check your package name


import android.content.Intent;

import android.content.IntentFilter;

import android.os.Bundle;

import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    private AirplaneModeReceiver airplaneModeReceiver;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the receiver
        airplaneModeReceiver = new AirplaneModeReceiver();
    }

    @Override
    protected void onStart() {
        super.onStart();
        // 1. Create an IntentFilter for Airplane Mode
        IntentFilter filter = new IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED);

        // 2. Register the receiver dynamically
        // This is required for Android 8.0 (Oreo) and above
        registerReceiver(airplaneModeReceiver, filter);
    }
```

```
    @Override
    protected void onStop() {
        super.onStop();
        // 3. Unregister the receiver to save battery and prevent crashes
        if (airplaneModeReceiver != null) {
            unregisterReceiver(airplaneModeReceiver);
        }
    }
}
```

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvInstruction"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Toggle Airplane Mode via your Quick Settings Panel to test."
```

```
        android:textSize="18sp"
        android:gravity="center"
        android:layout_marginBottom="20dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="(Toast notification will appear)"
        android:textSize="14sp"
        android:textColor="#666666"/>

</LinearLayout>
```

AirplaneModeReceiver.java

```
package com.example.airplanemodeapp;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.CountDownTimer;
import android.widget.Toast;

public class AirplaneModeReceiver extends BroadcastReceiver {
```

```
@Override
public void onReceive(Context context, Intent intent) {
    if (Intent.ACTION_AIRPLANE_MODE_CHANGED.equals(intent.getAction())) {

        boolean isAirplaneModeOn = intent.getBooleanExtra("state", false);
        String message = isAirplaneModeOn ? "Airplane Mode is ON      " : "Airplane Mode is OFF    ";

        // Call our custom helper method for a 10-second Toast
        showCustomDurationToast(context, message, 10000);
    }
}

// Helper method to keep a Toast visible for a specific duration (in milliseconds)
private void showCustomDurationToast(Context context, String message, int durationInMillis) {
    final Toast toast = Toast.makeText(context, message, Toast.LENGTH_SHORT);

    // Create a timer that updates the toast every 1 second (1000ms)
    new CountDownTimer(durationInMillis, 1000) {

        public void onTick(long millisUntilFinished) {
            // Keep showing the toast to prevent it from fading out
            toast.show();
        }
```

```
            public void onFinish() {
                // Ensure the toast disappears when time is up
                toast.cancel();
            }
        }.start();
    }
}
```

Q. Explain the Android Activity life cycle in detail. Implement a simple app that logs the lifecycle methods (onCreate(), onStart(), onResume(), onPause(), onStop(), onDestroy()).

Q. Write a program that logs the different lifecycle methods of an Activity (e.g., onCreate(), onStart(), onResume(), onPause(), onStop(), onDestroy()) to the logout

MainActivity.java

```
package com.example.lifecyclelogger; // TODO: Check your package name


import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    // Tag for filtering logs in Android Studio
```

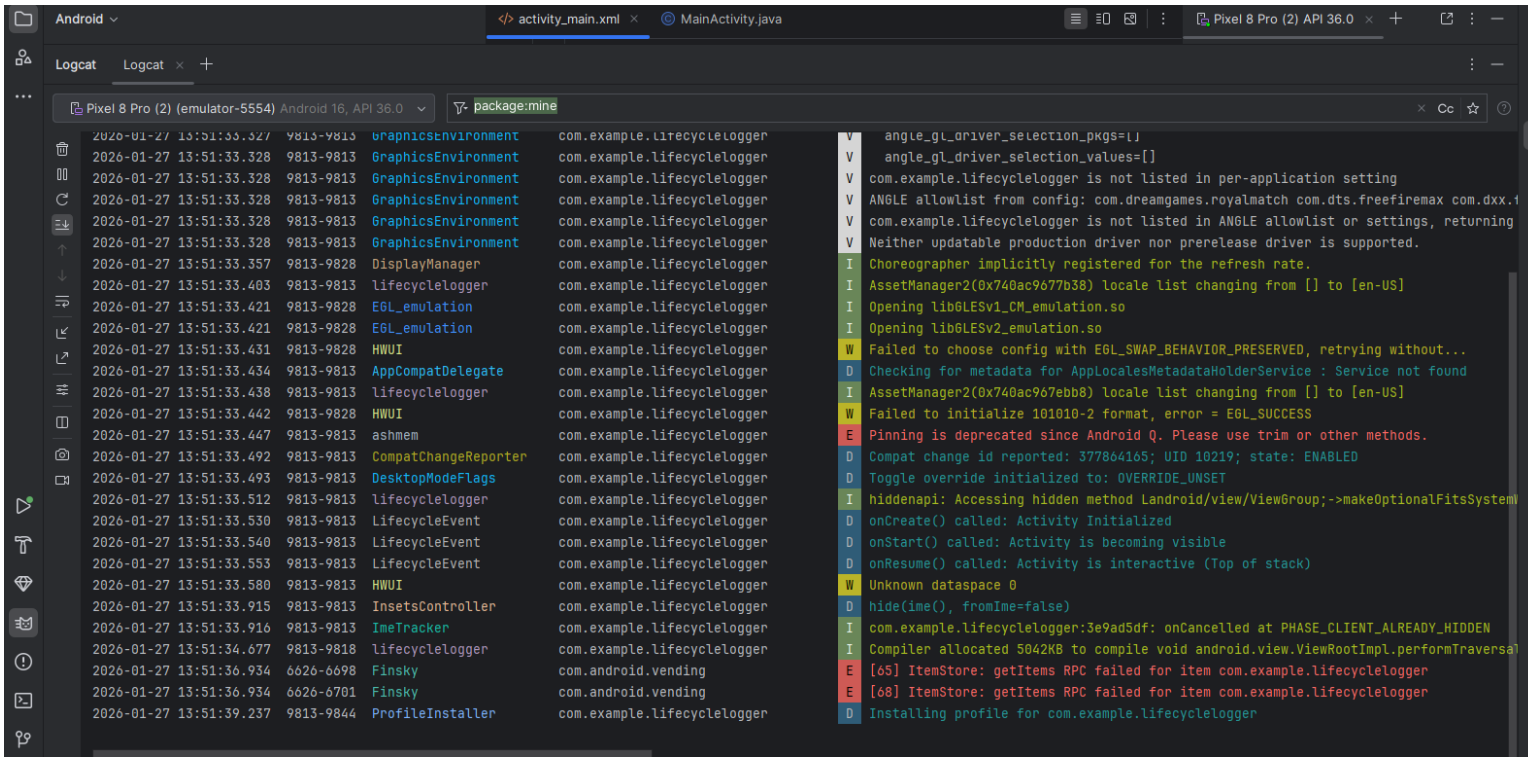
```
private static final String TAG = "LifecycleEvent";

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    Log.d(TAG, "onCreate() called: Activity Initialized");
}

@Override
protected void onStart() {
    super.onStart();
    Log.d(TAG, "onStart() called: Activity is becoming visible");
}

@Override
protected void onResume() {
    super.onResume();
    Log.d(TAG, "onResume() called: Activity is interactive (Top of stack)");
}
```

```
@Override
protected void onPause() {
    super.onPause();
    Log.d(TAG, "onPause() called: User is leaving (or app partially obscured)");
}

@Override
protected void onStop() {
    super.onStop();
    Log.d(TAG, "onStop() called: Activity is no longer visible");
}

@Override
protected void onRestart() {
    super.onRestart();
    Log.d(TAG, "onRestart() called: Activity returning from stopped state");
}

@Override
protected void onDestroy() {
```

```
        super.onDestroy();
        Log.d(TAG, "onDestroy() called: Activity is being destroyed");
    }
}
```

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Logcat for 'LifecycleEvent'"
        android:textSize="20sp"
        android:textStyle="bold" />
</LinearLayout>
```



Q. Create an android application to demonstrate the following event listeners on the following widgets:

Button: on click listener, long click listener

Image: Touch listener with these motion events :Action up, action down and action pointer down

Edit Text : key listener on Enter key

The Appropriate message should be displayed in text view.

Activity main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"

        android:orientation="vertical"

        android:padding="16dp"

        android:gravity="center_horizontal">


        <ImageView

            android:id="@+id/eventImage"

            android:layout_width="200dp"

            android:layout_height="200dp"

            android:layout_marginTop="100dp"

            android:background="#DDDDDD"

            android:contentDescription="Touch Area"

            android:src="@mipmap/ic_launcher" />

        <Button

            android:id="@+id/eventButton"

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:text="Click or Long Click Me" />


        <TextView

            android:id="@+id/statusTextView"

            android:layout_width="match_parent"

            android:layout_height="76dp"

            android:layout_marginBottom="20dp"

            android:gravity="center"
```

```
        android:text="Event Status will appear here"
        android:textColor="#333333"
        android:textSize="18sp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/eventEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:hint="Type and press Enter"
        android:inputType="text" />

</LinearLayout>
```

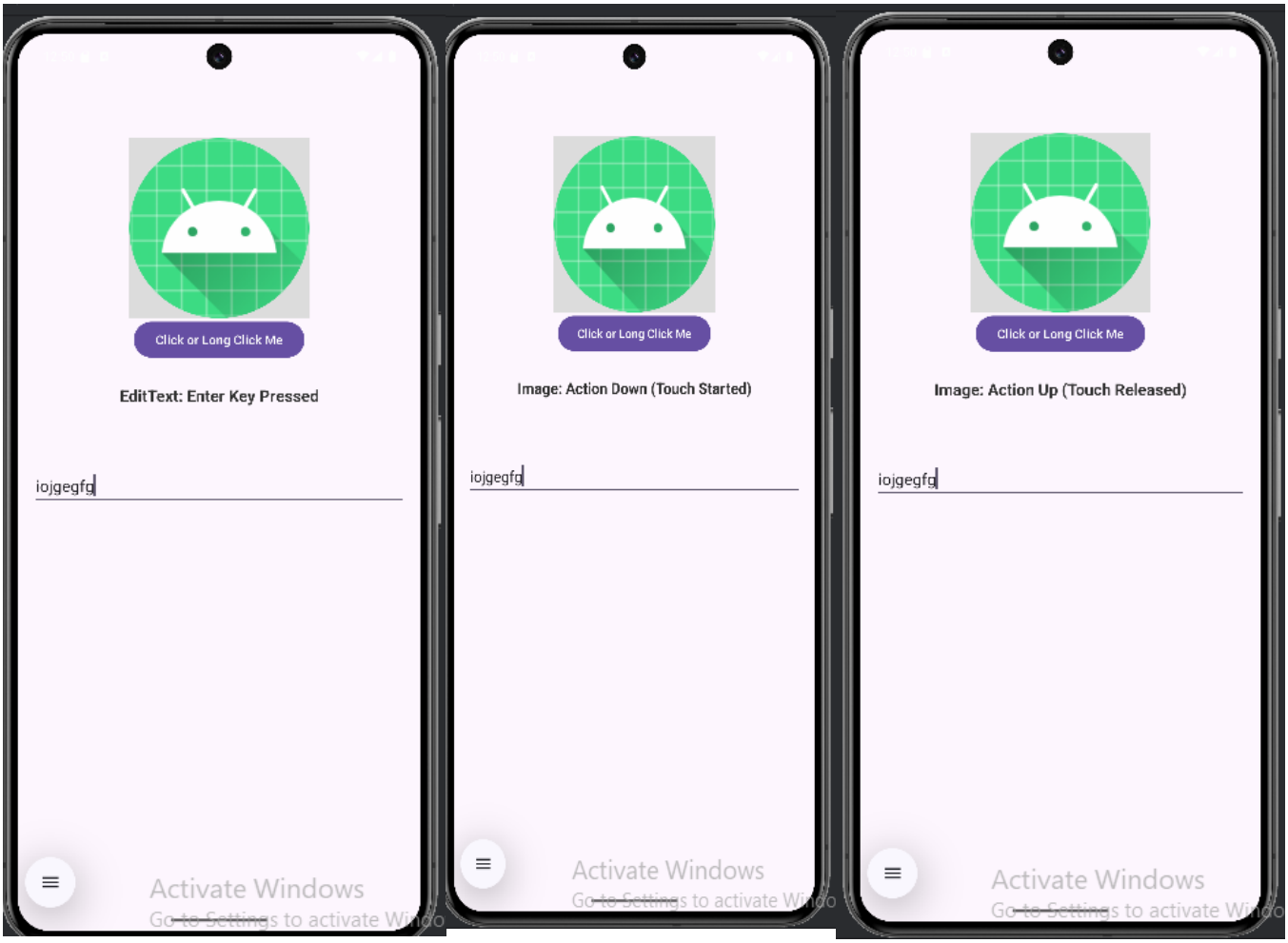
Main activity.java

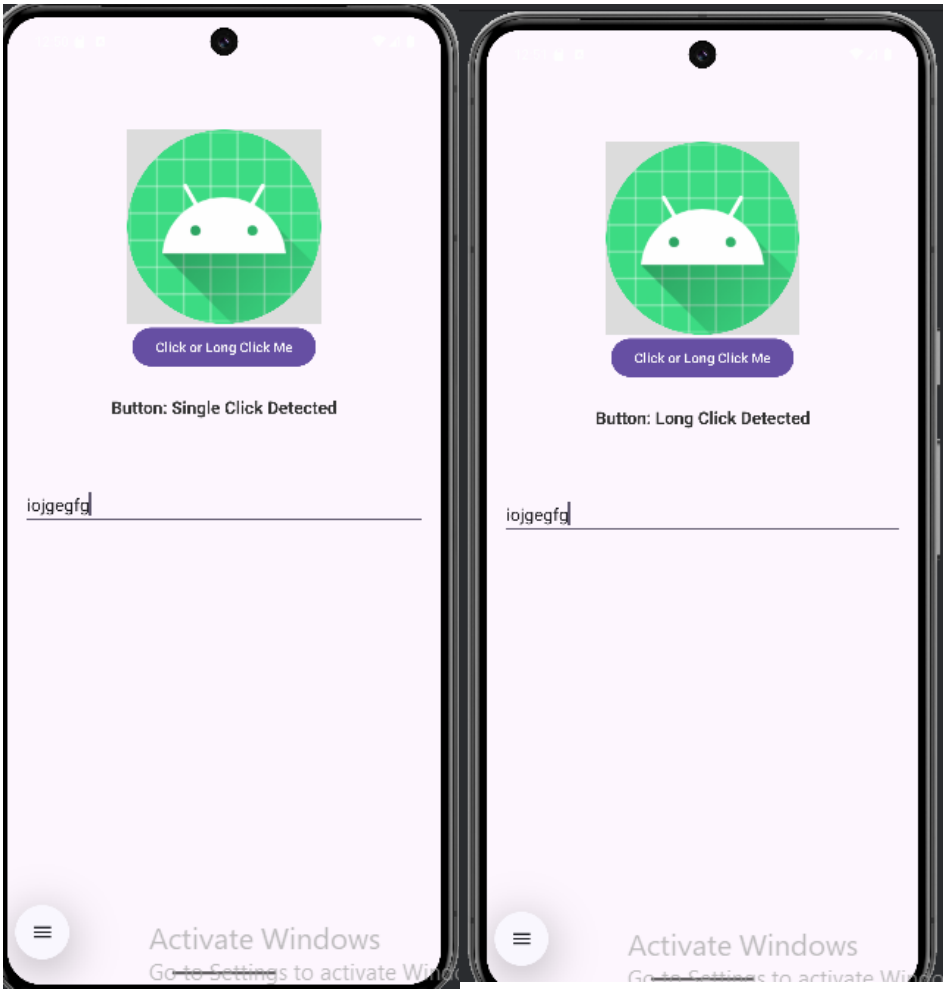
```
package com.example.eventlistenerapp;

import android.os.Bundle;
import android.view.KeyEvent;
import android.view.MotionEvent;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ImageView;
import android.widget.TextView;
```

```java
import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    private TextView statusTextView;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        statusTextView = findViewById(R.id.statusTextView);

        Button btn = findViewById(R.id.eventButton);

        ImageView img = findViewById(R.id.eventImage);

        EditText editText = findViewById(R.id.eventEditText);


        // --- 1. Button Listeners ---

        btn.setOnClickListener(v -> statusTextView.setText("Button: Single Click
Detected"));


        btn.setOnLongClickListener(v -> {

            statusTextView.setText("Button: Long Click Detected");

            return true; // Return true to consume the event

        });


        // --- 2. Image Touch Listener (Motion Events) ---
```

```
        img.setOnTouchListener((v, event) -> {
            int action = event.getActionMasked();

            switch (action) {
                case MotionEvent.ACTION_DOWN:
                    statusTextView.setText("Image: Action Down (Touch Started)");
                    break;
                case MotionEvent.ACTION_UP:
                    statusTextView.setText("Image: Action Up (Touch Released)");
                    break;
                case MotionEvent.ACTION_POINTER_DOWN:
                    statusTextView.setText("Image: Action Pointer Down (Multi-touch)");
                    break;
            }
            return true; // Consuming the touch event
        });

        // --- 3. EditText Key Listener (Enter Key) ---
        editText.setOnKeyListener((v, keyCode, event) -> {
            // We check for ACTION_DOWN so it only triggers once per press
            if (event.getAction() == KeyEvent.ACTION_DOWN && keyCode ==
KeyEvent.KEYCODE_ENTER) {
                statusTextView.setText("EditText: Enter Key Pressed");
                return true;
            }
            return false;
```

```
        });

    }

}
```


Click or Long Click Me
EditText: Enter Key Pressed
iojgegfg
Click or Long Click Me
Image: Action Down (Touch Started)
iojgegfg
Click or Long Click Me
Image: Action Up (Touch Released)
iojgegfg
Activate Windows
Go to Settings to activate Windows



Q. Implement a custom color scheme for your registration form. Set the background color of the entire layout to a light color, and use a contrasting color for the Submit button.

Colors.xml

```
<?xml version="1.0" encoding="utf-8"?>

<resources>

    <color name="form_background_light">#F8F9FA</color>
```

```
    <color name="submit_button_primary">#1A73E8</color>


    <color name="white">#FFFFFF</color>
</resources>
```

Activitymain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main_layout_id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_marginTop="100dp"
    android:background="@color/form_background_light"
    android:gravity="center_horizontal">


    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="User Registration"
        android:textSize="24sp"
        android:layout_marginBottom="30dp"
        android:textColor="#202124" />
```

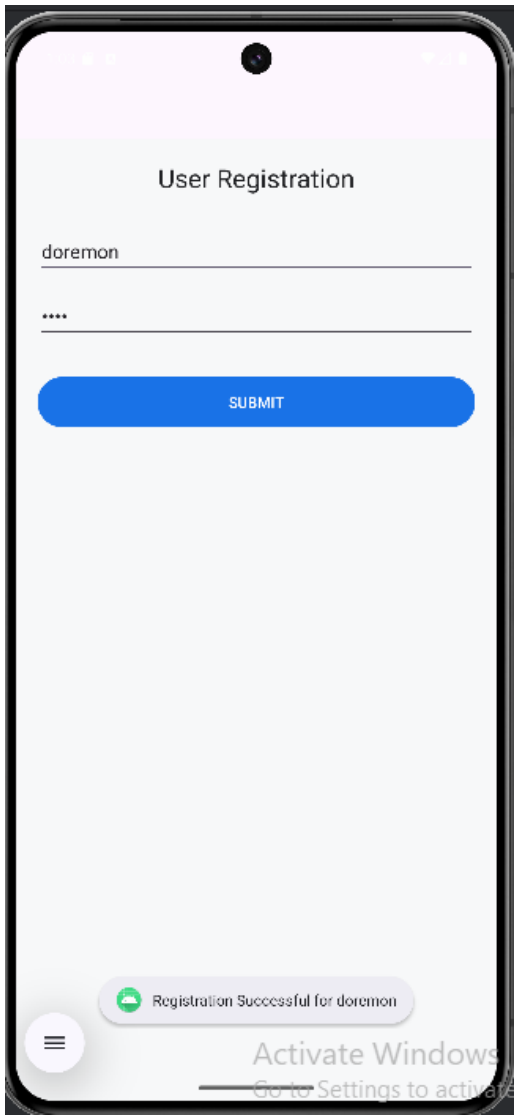
```
    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:layout_marginBottom="15dp" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginBottom="30dp" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="55dp"
        android:text="SUBMIT"
        android:textColor="@color/white"
        android:backgroundTint="@color/submit_button_primary" />

</LinearLayout>
```

Main Activity.java

```java
package com.example.registerationformcolor;


import android.graphics.Color;

import android.os.Bundle;

import android.widget.Button;

import android.widget.EditText;

import android.widget.LinearLayout;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    private EditText etUsername, etPassword;

    private Button btnSubmit;

    private LinearLayout rootLayout;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        // Initialize Views

        rootLayout = findViewById(R.id.main_layout_id); // Ensure this ID exists in your XML root

        etUsername = findViewById(R.id.etUsername);
```

```java
        etPassword = findViewById(R.id.etPassword);
        btnSubmit = findViewById(R.id.btnSubmit);

        // OPTIONAL: Set colors programmatically
        // (Use this if you want to override XML values dynamically)
        applyCustomColors();

        // Handle the Submit Action
        btnSubmit.setOnClickListener(v -> {
            String name = etUsername.getText().toString();
            if (name.isEmpty()) {
                Toast.makeText(this, "Please enter a name", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Registration Successful for " + name,
Toast.LENGTH_LONG).show();
            }
        });
    }

    /**
     * Demonstrates how to apply the light background and
     * contrasting button color using Java code.
     */
    private void applyCustomColors() {
        // Setting the light background (Off-white)
        int lightColor = ContextCompat.getColor(this, R.color.form_background_light);
```

```
        rootLayout.setBackgroundColor(lightColor);


        // Setting the contrasting button color
        int contrastColor = ContextCompat.getColor(this, R.color.submit_button_primary);
        btnSubmit.setBackgroundColor(contrastColor);
        btnSubmit.setTextColor(Color.WHITE);
    }
}
```



Q. **Insert the new contents in the following resources and demonstrate their uses in the android application Android Resources: (Color, Theme, String, Drawable, Dimension, Image).**

******DONE IN PRAC 1******

Q. Define the following string resources in the strings.xml file for the registration form:

Full Name, Email Password Confirm Password and Submit
Use these string resources in the corresponding TextView and Button in your layout XML.

Q. Create a registration form where the form fields (Name, Email, Password, Confirm Password) are stacked vertically. Include a Submit button at the bottom of the form.

res/values/strings.xml

```
<resources>
   <string name="app_name">RegistrationApp</string>

   <string name="label_full_name">Full Name</string>
   <string name="label_email">Email</string>
   <string name="label_password">Password</string>
   <string name="label_confirm_password">Confirm Password</string>
   <string name="btn_submit">Submit</string>
</resources>
```

Activity_main.xml

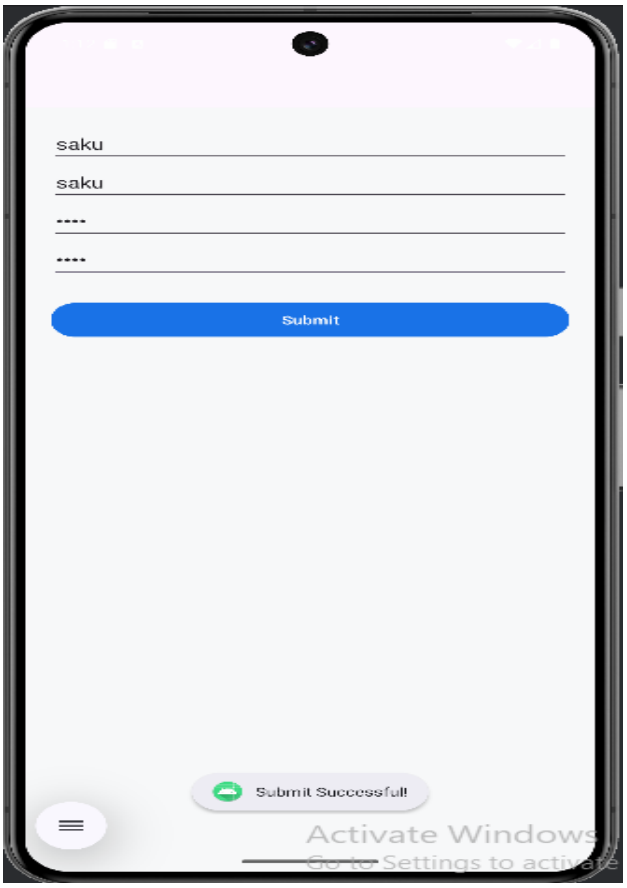
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main_layout_id"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp"
   android:background="@color/form_background_light"
   android:layout_marginTop="100dp"
   >

   <EditText
       android:id="@+id/etFullName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="@string/label_full_name"
       android:inputType="textPersonName" />
```

```
    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/label_email"
        android:inputType="textEmailAddress" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/label_password"
        android:inputType="textPassword" />

    <EditText
        android:id="@+id/etConfirmPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/label_confirm_password"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="@string/btn_submit"
        android:backgroundTint="@color/submit_button_primary"
        android:textColor="@color/white" />

</LinearLayout>
```

MainActivity.java

```
package com.example.registerationformcolor;

import android.graphics.Color;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.LinearLayout;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

    private EditText etUsername, etPassword;
    private Button btnSubmit;
```

```java
    private LinearLayout rootLayout;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 1. MUST Initialize Views (Link Java to XML)
        rootLayout = findViewById(R.id.main_layout_id);
        etUsername = findViewById(R.id.etFullName); // Use the ID you set in XML
        etPassword = findViewById(R.id.etPassword);
        btnSubmit = findViewById(R.id.btnSubmit);

        // 2. Call the method to apply colors
        applyCustomColors();

        // 3. Handle the Submit Action
        btnSubmit.setOnClickListener(v -> {
            String successMsg = getString(R.string.btn_submit) + " Successful!";
            Toast.makeText(this, successMsg, Toast.LENGTH_SHORT).show();
        });
    }

    /**
     * Applies the light background and contrasting button color.
     */
    private void applyCustomColors() {
        // Double-check: Ensure these IDs exist in colors.xml
        int lightColor = ContextCompat.getColor(this, R.color.form_background_light);
        rootLayout.setBackgroundColor(lightColor);

        int contrastColor = ContextCompat.getColor(this, R.color.submit_button_primary);
        btnSubmit.setBackgroundColor(contrastColor);
        btnSubmit.setTextColor(Color.WHITE);
    }
}
```



Q. Create an android application, with one button and image view also set the background image in the application, the image should be changed on each click of the button (Use two images).

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/main_root_layout"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:gravity="center"

   android:padding="20dp"
```

```
    android:background="@drawable/bg_app"> <TextView

    android:id="@+id/tvStatus"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Current Image: One"

    android:textSize="22sp"

    android:textStyle="bold"

    android:textColor="#FFFFFF"

    android:layout_marginBottom="20dp"/>


    <ImageView

        android:id="@+id/ivDisplay"

        android:layout_width="250dp"

        android:layout_height="250dp"

        android:src="@drawable/image_one"

        android:scaleType="centerCrop"

        android:layout_marginBottom="30dp"

        android:background="#80000000"/> <Button

        android:id="@+id/btnChange"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Change Image"

        android:padding="15dp"/>


</LinearLayout>
```

MainActivity.java

```
package com.example.imagechanger;


import android.os.Bundle;

import android.widget.Button;

import android.widget.ImageView;

import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


   private TextView tvStatus;

   private ImageView ivDisplay;

   private Button btnChange;

   // Flag to track which image is currently showing

   private boolean isImageOne = true;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // 1. Initialize Views

       tvStatus = findViewById(R.id.tvStatus);

       ivDisplay = findViewById(R.id.ivDisplay);
```

```
        btnChange = findViewById(R.id.btnChange);


        // 2. Set Button Click Listener using a Lambda (cleaner than anonymous class)
        btnChange.setOnClickListener(v -> {
            if (isImageOne) {
                // Switch to second image
                ivDisplay.setImageResource(R.drawable.image_two);
                tvStatus.setText("Current Image: Two");
            } else {
                // Switch back to first image
                ivDisplay.setImageResource(R.drawable.image_one);
                tvStatus.setText("Current Image: One");
            }

            // Toggle the flag
            isImageOne = !isImageOne;
        });
    }
}
```

Q. Create an android application to demonstrate the use of sub menu the toast should be appeared by selecting the sub menu item

Themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">

   <!-- Base application theme. -->

   <style name="Base.Theme.Submenuapp" parent="Theme.AppCompat.Light.DarkActionBar">

       <!-- Customize your light theme here. -->

       <!-- <item name="colorPrimary">@color/my_light_primary</item> -->

   </style>
```

```
    <style name="Theme.Submenuapp" parent="Base.Theme.Submenuapp" />

</resources>
```

The Menu Resource (res/menu/main_menu.xml)

First, ensure you have a menu folder in your res directory. This file defines the "parent" item and the nested "sub-menu" items.

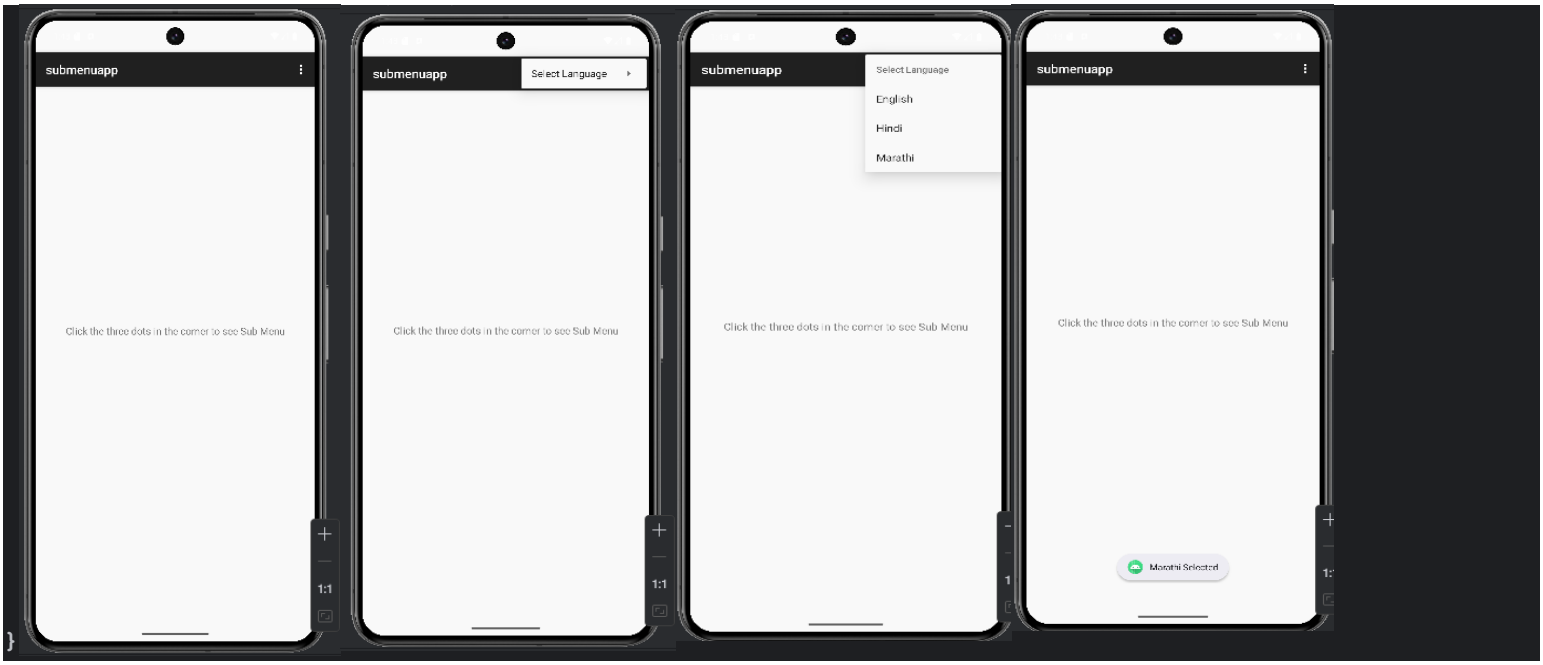
```
<?xml version="1.0" encoding="utf-8"?>

<menu xmlns:android="http://schemas.android.com/apk/res/android">



    <item

        android:id="@+id/menu_languages"

        android:title="Select Language">



        <menu>

            <item

                android:id="@+id/sub_english"

                android:title="English" />

            <item

                android:id="@+id/sub_hindi"

                android:title="Hindi" />

            <item

                android:id="@+id/sub_marathi"

                android:title="Marathi" />

        </menu>

    </item>
```

```
</menu>
```

Mainactivity.java

```
package com.example.submenuapp;


import android.os.Bundle;

import android.view.Menu;

import android.view.MenuInflater;

import android.view.MenuItem;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

    }


    // THIS METHOD MAKES THE MENU APPEAR

    @Override

    public boolean onCreateOptionsMenu(Menu menu) {

        MenuInflater inflater = getMenuInflater();

        // This 'main_menu' refers to your res/menu/main_menu.xml file

        inflater.inflate(R.menu.main_menu, menu);
```

```
        return true;
    }

    // THIS METHOD HANDLES THE SUB-MENU CLICKS
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        // Matching IDs from your main_menu.xml screenshot
        if (id == R.id.sub_english) {
            Toast.makeText(this, "English Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.sub_hindi) {
            Toast.makeText(this, "Hindi Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.sub_marathi) {
            Toast.makeText(this, "Marathi Selected", Toast.LENGTH_SHORT).show();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
```



Q. Design a form with a RadioGroup for selecting the gender (Male/Female/Other). Include a Submit button that shows the selected gender in a TextView.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:layout_marginTop="100dp"

   android:padding="20dp"

   android:gravity="center_horizontal">


   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"
```

```
        android:text="Select Gender:"
        android:textSize="18sp"
        android:layout_marginBottom="10dp" />

    <RadioGroup
        android:id="@+id/genderRadioGroup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">

        <RadioButton
            android:id="@+id/rbMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male" />

        <RadioButton
            android:id="@+id/rbFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female" />

        <RadioButton
            android:id="@+id/rbOther"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Other" />
```

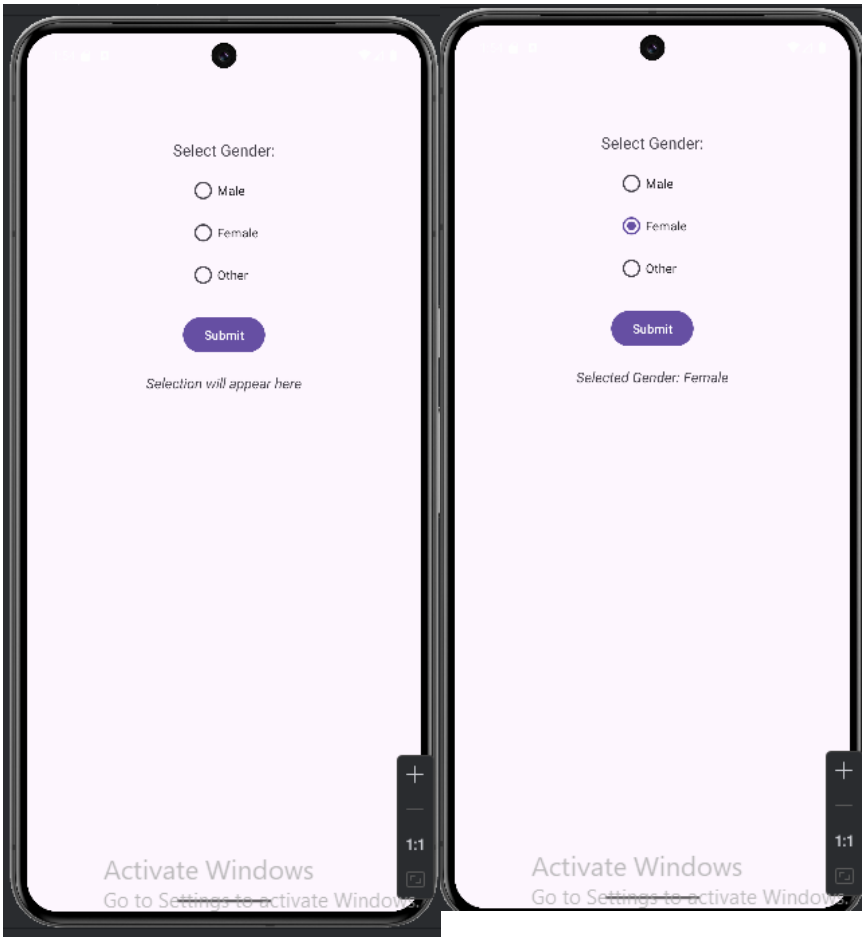
```
    </RadioGroup>


    <Button
        android:id="@+id/btnSubmitGender"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_marginTop="20dp" />


    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Selection will appear here"
        android:textSize="16sp"
        android:layout_marginTop="20dp"
        android:textStyle="italic" />


</LinearLayout>
```

MainActivity

```
package com.example.genderform; // Ensure this matches your package name


import android.os.Bundle;
import android.widget.Button;
import android.widget.RadioButton;
```

```java
import android.widget.RadioGroup;

import android.widget.TextView;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    private RadioGroup genderRadioGroup;

    private Button btnSubmit;

    private TextView tvResult;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        // Initialize Views

        genderRadioGroup = findViewById(R.id.genderRadioGroup);

        btnSubmit = findViewById(R.id.btnSubmitGender);

        tvResult = findViewById(R.id.tvResult);


        btnSubmit.setOnClickListener(v -> {

            // 1. Get the ID of the selected RadioButton

            int selectedId = genderRadioGroup.getCheckedRadioButtonId();


            if (selectedId == -1) {
```

```
            // No radio button was checked
            Toast.makeText(this, "Please select a gender", Toast.LENGTH_SHORT).show();
        } else {
            // 2. Find the RadioButton by the ID
            RadioButton selectedRadioButton = findViewById(selectedId);

            // 3. Get the text of that RadioButton
            String gender = selectedRadioButton.getText().toString();

            // 4. Update the TextView
            tvResult.setText("Selected Gender: " + gender);
        }
    });
    }
}
```



Q. Create a RelativeLayout for a profile screen with:

A Profile Image centered at the top of the screen.

A TextView displaying the name below the profile image, aligned to the center.

A Button below the name, aligned to the center.

Activity_main

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="16dp">

   <ImageView
       android:id="@+id/ivProfile"
       android:layout_width="150dp"
       android:layout_height="150dp"
       android:layout_centerInParent="true"
       android:src="@drawable/image_one"
       android:contentDescription="Profile Picture" />
```

```
    <TextView
        android:id="@+id/tvProfileName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/ivProfile"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="16dp"
        android:text="John Doe"
        android:textSize="24sp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/btnEditProfile"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/tvProfileName"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:text="Edit Profile" />

</RelativeLayout>
```

MainAcivity

```
package com.example.profilescreen;

import android.os.Bundle;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ImageView ivProfile = findViewById(R.id.ivProfile);
        TextView tvName = findViewById(R.id.tvProfileName);
        Button btnEdit = findViewById(R.id.btnEditProfile);

        btnEdit.setOnClickListener(v -> {
            Toast.makeText(this, "Opening Profile Editor...", Toast.LENGTH_SHORT).show();
        });
    }
}
```

Q. Design a ListView that displays a list of items. Each item should be a simple TextView with a string from an array of names. The list should be populated dynamically in your Activity.

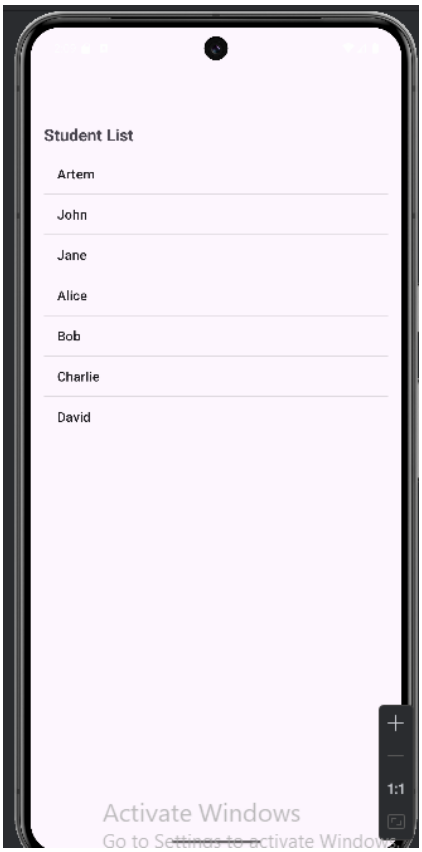
Activity_main

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_marginTop="100dp">

   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Student List"
       android:textSize="20sp"
       android:textStyle="bold"
       android:layout_marginBottom="10dp"/>

   <ListView
       android:id="@+id/nameListView"
       android:layout_width="match_parent"
       android:layout_height="wrap_content" />

</LinearLayout>
```

MainActivity

```
package com.example.listviewapp; // Ensure this matches your project package

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   // Data Source
   String[] names = {"Artem", "John", "Jane", "Alice", "Bob", "Charlie", "David"};

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       // 1. Initialize the ListView
       ListView listView = findViewById(R.id.nameListView);
```

```
        // 2. Initialize the Adapter
        // Parameters: Context, Layout for each row, Data source
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                names
        );

        // 3. Set the adapter to the ListView
        listView.setAdapter(adapter);

        // 4. Handle item clicks (Optional but recommended)
        listView.setOnItemClickListener((parent, view, position, id) -> {
            String selectedName = names[position];
            Toast.makeText(this, "Clicked: " + selectedName, Toast.LENGTH_SHORT).show();
        });
    }
}
```



Q. Create an Activity that hosts a Fragment. When a button in the Activity is clicked, pass a string value to the Fragment and display it in a TextView.

DisplayFragment.java

```
package com.example.fragmentapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class DisplayFragment extends Fragment {

   @Nullable
   @Override
   public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup container,
@Nullable Bundle savedInstanceState) {
       View view = inflater.inflate(R.layout.fragment_display, container, false);
       TextView tvDisplay = view.findViewById(R.id.tvFragmentDisplay);

       // Retrieve the data passed from the activity
       if (getArguments() != null) {
           String receivedData = getArguments().getString("msg_key");
           tvDisplay.setText(receivedData);
       }

       return view;
   }
}
```

MainActivity.java

```
package com.example.fragmentapp;

import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       Button btnSend = findViewById(R.id.btnSendToFragment);

       btnSend.setOnClickListener(v -> {
           // 1. Create the Fragment instance
```

```
            DisplayFragment fragment = new DisplayFragment();

            // 2. Prepare the data in a Bundle
            Bundle bundle = new Bundle();
            bundle.putString("msg_key", "Hello from Activity!");
            fragment.setArguments(bundle);

            // 3. Perform the Fragment Transaction
            getSupportFragmentManager().beginTransaction()
                    .replace(R.id.fragment_container, fragment)
                    .commit();
        });
    }
}
```

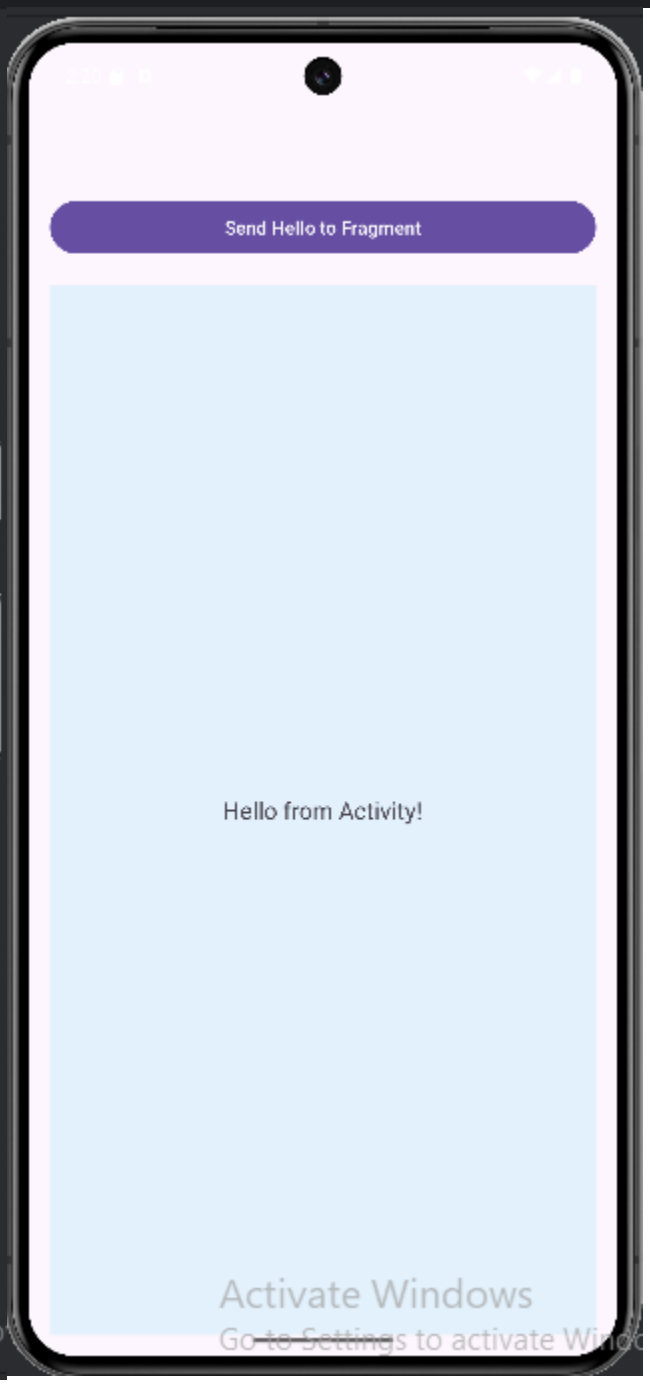
Fragment_display.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:background="#E3F2FD">

    <TextView
        android:id="@+id/tvFragmentDisplay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Waiting for data..."
        android:textSize="18sp" />
</LinearLayout>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_marginTop="100dp">

    <Button
        android:id="@+id/btnSendToFragment"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send Hello to Fragment" />

    <FrameLayout
```

```
        android:id="@+id/fragment_container"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="20dp" />

</LinearLayout>
```



- **Create an android application using linear layout and insert 10 animals in the list view and display the appropriate Toast.**

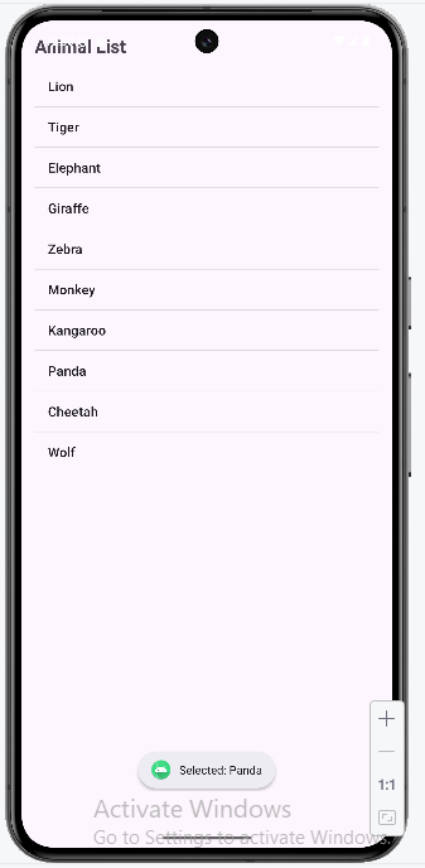
Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">

  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Animal List"
     android:textSize="22sp"
     android:textStyle="bold"
     android:layout_marginBottom="10dp"/>

  <ListView
     android:id="@+id/animalListView"
     android:layout_width="match_parent"
     android:layout_height="wrap_content" />

</LinearLayout>
```

MainActivity.java

```
package com.example.animallistapp;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  // 1. Data Source: 10 Animals
  String[] animals = {
        "Lion", "Tiger", "Elephant", "Giraffe", "Zebra",
        "Monkey", "Kangaroo", "Panda", "Cheetah", "Wolf"
```

```
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 2. Initialize ListView
        ListView listView = findViewById(R.id.animalListView);

        // 3. Create the Adapter
        // Uses simple_list_item_1, a built-in layout for a single TextView row
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                animals
        );

        // 4. Set the Adapter to the ListView
        listView.setAdapter(adapter);

        // 5. Set Item Click Listener to display Toast
        listView.setOnItemClickListener((parent, view, position, id) -> {
            String selectedAnimal = animals[position];
            Toast.makeText(MainActivity.this, "Selected: " + selectedAnimal, Toast.LENGTH_SHORT).show();
        });
    }
}
```



- **Create an app that displays an ImageView. When the user clicks the ImageView, show a Toast with the message & Image clicke**d!

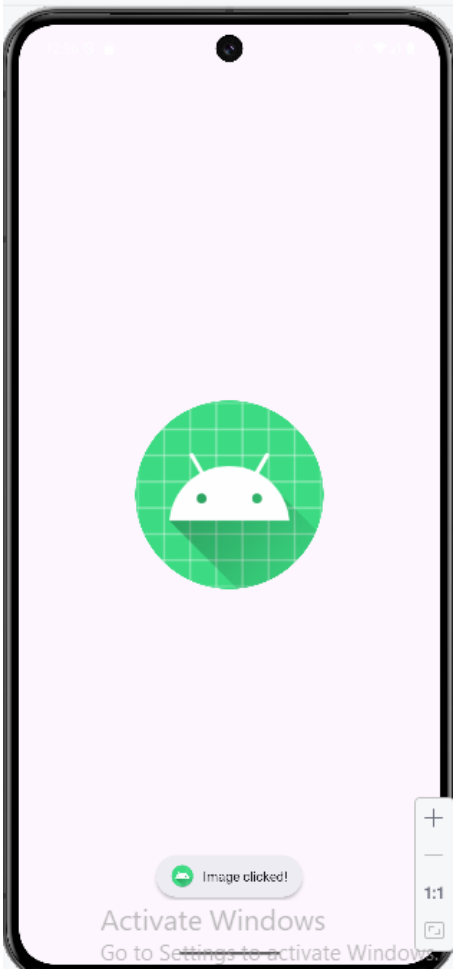
Activity main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical">

   <ImageView
      android:id="@+id/myImageView"
      android:layout_width="200dp"
      android:layout_height="200dp"
      android:src="@mipmap/ic_launcher"
      android:contentDescription="Clickable Image"
      android:clickable="true"
      android:focusable="true" />
```

```
</LinearLayout>
```

MainActivity.jaava

```
package com.example.imageclickapp; // Double-check this matches your package name

import android.os.Bundle;
import android.widget.ImageView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      // 1. Initialize the ImageView
      ImageView imageView = findViewById(R.id.myImageView);

      // 2. Set the Click Listener
      imageView.setOnClickListener(v -> {
          // 3. Display the Toast message
          Toast.makeText(MainActivity.this, "Image clicked!", Toast.LENGTH_SHORT).show();
      });
  }
}
```



- Design a simple login form using ConstraintLayout that has:

  Username and Password fields aligned vertically.

  Login button centered horizontally below the fields.

  Forgot Password? link aligned to the right of the password field.

  Task: Position all the elements using constraints in the ConstraintLayout.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="24dp">
```

```
<EditText
    android:id="@+id/etUsername"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="276dp"
    android:hint="Username"
    android:inputType="textPersonName"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/etPassword"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:hint="Password"
    android:inputType="textPassword"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/etUsername" />

<TextView
    android:id="@+id/tvForgotPassword"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Forgot Password?"
    android:textColor="#3F51B5"
    android:textSize="14sp"
    android:layout_marginTop="8dp"
    app:layout_constraintTop_toBottomOf="@id/etPassword"
    app:layout_constraintEnd_toEndOf="@id/etPassword" />

<Button
    android:id="@+id/btnLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
    android:layout_marginTop="40dp"
    app:layout_constraintTop_toBottomOf="@id/tvForgotPassword"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity

```
package com.example.loginform;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    EditText etUsername = findViewById(R.id.etUsername);
    EditText etPassword = findViewById(R.id.etPassword);
    Button btnLogin = findViewById(R.id.btnLogin);
    TextView tvForgot = findViewById(R.id.tvForgotPassword);

    btnLogin.setOnClickListener(v -> {
      String user = etUsername.getText().toString();
      Toast.makeText(this, "Attempting login for: " + user, Toast.LENGTH_SHORT).show();
    });
  }
}
```

- **Design a screen with two buttons: Start and Stop. When the Start button is clicked, start a Timer (display the time in seconds). When the Stop button is clicked, stop the timer.**

acitivity_main

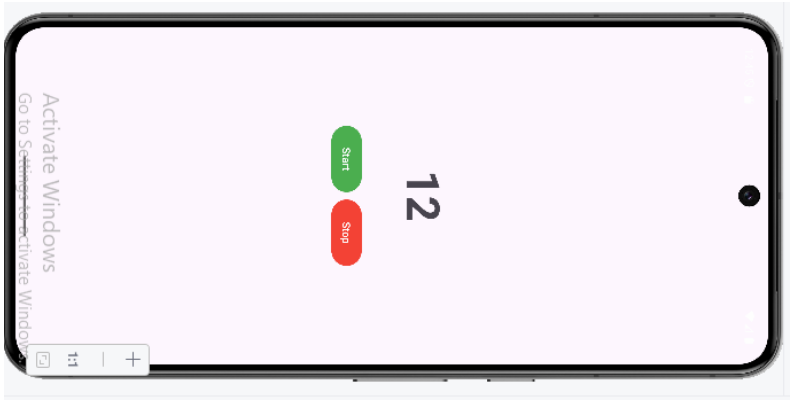
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical"
```

```
    android:padding="20dp">

    <TextView
        android:id="@+id/tvTimer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0"
        android:textSize="60sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:orientation="horizontal">

        <Button
            android:id="@+id/btnStart"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Start"
            android:layout_marginEnd="10dp"
            android:backgroundTint="#4CAF50" />

        <Button
            android:id="@+id/btnStop"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Stop"
            android:backgroundTint="#F44336" />
    </LinearLayout>

</LinearLayout>
```

MainActivity

```
package com.example.timerapp; // Ensure this matches your project package

import android.os.Bundle;
import android.os.Handler;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    private TextView tvTimer;
    private int seconds = 0;
    private boolean running = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvTimer = findViewById(R.id.tvTimer);
        Button btnStart = findViewById(R.id.btnStart);
        Button btnStop = findViewById(R.id.btnStop);

        // Start button logic
        btnStart.setOnClickListener(v -> {
            running = true;
        });

        // Stop button logic
        btnStop.setOnClickListener(v -> {
            running = false;
        });

        // Run the timer logic
        runTimer();
    }

    private void runTimer() {
        final Handler handler = new Handler();
```




```
handler.post(new Runnable() {
    @Override
    public void run() {
        // Update the TextView
        tvTimer.setText(String.valueOf(seconds));

        // If running is true, increment the seconds
        if (running) {
            seconds++;
        }

        // Post the code again with a delay of 1 second
        handler.postDelayed(this, 1000);
```

```
        }
    });
  }
}
```

Combined

- Create a simple Registration Form with the following fields:

  Name (Text Input)

  Email (Text Input)

  Phone Number (Text Input)

  Password (Password Input)

  Confirm Password (Password Input)

  Submit Button

- Create a Registration Form with the following fields:

  First Name (Text Input)

  Last Name (Text Input)

  Email Address (Text Input)

  Country (Spinner Dropdown)

  Password (Password Input)

  Confirm Password (Password Input)

  Submit Button

- Create a Registration Form with the following fields:

  Username (Text Input)

  Email Address (Text Input)

  Gender (Radio Buttons with options ;Male and Female

Password (Password Input)

Submit Button

- Create a Registration Form with the following fields:

  Username (Text Input)

  Email Address (Text Input)

  Phone Number (Text Input)

  Password (Password Input)

  Submit Button (Floating Action Button)

- Design a Registration Form that includes:

  Name (Text Input)

  Date of Birth (Date Picker)

  Email Address (Text Input)

  Phone Number (Text Input)

  Password (Password Input)

  Submit Button

- Create a Registration Form with the following fields:

  Username (Text Input)

  Email Address (Text Input)

  Agree to Terms and Conditions (Checkbox)

Password (Password Input)

Confirm Password (Password Input)

Submit Button

- Create a Registration Form with the following fields:

  Username (Text Input)

  Email Address (Text Input)

  Password (Password Input)

  Confirm Password (Password Input)

  Profile Picture (Image Upload Button)

  Submit Button

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#F5F7FA">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="User Registration"
      android:textSize="24sp"
      android:textStyle="bold"
```

```
    android:layout_gravity="center"
    android:layout_marginBottom="20dp"
    android:textColor="#333"/>

<ImageView
    android:id="@+id/ivProfile"
    android:layout_width="100dp"
    android:layout_height="100dp"
    android:layout_gravity="center"
    android:src="@mipmap/ic_launcher_round"
    android:background="@drawable/image_border"
    android:scaleType="centerCrop"
    android:layout_marginBottom="5dp"/>

<Button
    android:id="@+id/btnUploadImage"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:text="Upload Photo"
    android:textSize="10sp"
    android:layout_marginBottom="10dp"
    style="@style/Widget.AppCompat.Button.Borderless.Colored"/>

<EditText
    android:id="@+id/etFullName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Full Name"
    android:inputType="textPersonName" />

<EditText
    android:id="@+id/etEmail"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Email"
    android:inputType="textEmailAddress" />

<EditText
    android:id="@+id/etPhone"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Phone Number"
    android:inputType="phone" />
```

```
    <Button
        android:id="@+id/btnDatePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Date of Birth"
        android:layout_marginTop="10dp" />

    <TextView
        android:id="@+id/tvDob"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="DOB: Not Selected"
        android:padding="5dp" />

    <RadioGroup
        android:id="@+id/rgGender"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="10dp">
        <RadioButton android:id="@+id/rbMale" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Male" />
        <RadioButton android:id="@+id/rbFemale" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Female" />
    </RadioGroup>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Country:"
        android:layout_marginTop="10dp"/>
    <Spinner
        android:id="@+id/spinnerCountry"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />
```

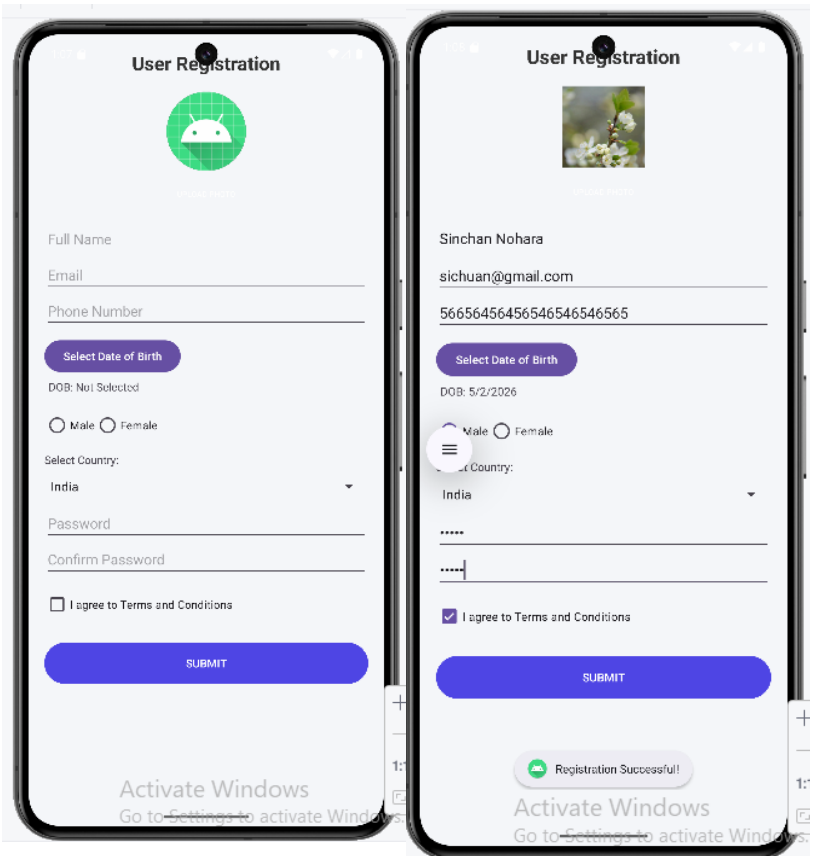
```
    <EditText
        android:id="@+id/etConfirmPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Confirm Password"
        android:inputType="textPassword" />

    <CheckBox
        android:id="@+id/cbTerms"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="I agree to Terms and Conditions"
        android:layout_marginTop="10dp" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="60dp"
        android:text="SUBMIT"
        android:layout_marginTop="20dp"
        android:backgroundTint="#4F46E5"
        android:textColor="#FFF" />

  </LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.registrationform;
import android.app.DatePickerDialog;
import android.net.Uri;
import android.os.Bundle;
import android.widget.*;
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

  private EditText etName, etEmail, etPass, etConfirm, etPhone;
  private Spinner spinnerCountry;
  private RadioGroup rgGender;
  private CheckBox cbTerms;
  private TextView tvDob;
```

```
    private Button btnSubmit, btnDatePicker;
    private ImageView ivProfile;
    // 1. Define the Image Picker Launcher
    private final ActivityResultLauncher<String> mGetContent = registerForActivityResult(
            new ActivityResultContracts.GetContent(),
            uri -> {
                if (uri != null) {
                    ivProfile.setImageURI(uri);
                    Toast.makeText(this, "Profile Picture Updated", Toast.LENGTH_SHORT).show();
                }
            });
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 2. Initialize all views
        ivProfile = findViewById(R.id.ivProfile);
        etName = findViewById(R.id.etFullName);
        etEmail = findViewById(R.id.etEmail);
        etPhone = findViewById(R.id.etPhone);
        etPass = findViewById(R.id.etPassword);
        etConfirm = findViewById(R.id.etConfirmPassword);
        rgGender = findViewById(R.id.rgGender);
        cbTerms = findViewById(R.id.cbTerms);
        tvDob = findViewById(R.id.tvDob);
        btnSubmit = findViewById(R.id.btnSubmit);
        btnDatePicker = findViewById(R.id.btnDatePicker);
        spinnerCountry = findViewById(R.id.spinnerCountry);
        // 3. Set Image Click Listener to trigger "Upload"
        ivProfile.setOnClickListener(v -> mGetContent.launch("image/*"));
        // 4. Setup Spinner for Countries
        String[] countries = {"India", "USA", "UK", "Canada", "Australia"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this, android.R.layout.simple_spinner_dropdown_item,
countries);
        spinnerCountry.setAdapter(adapter);
        // 5. Date Picker Logic
        btnDatePicker.setOnClickListener(v -> {
            Calendar c = Calendar.getInstance();
            new DatePickerDialog(this, (view, year, month, dayOfMonth) ->
                    tvDob.setText("DOB: " + dayOfMonth + "/" + (month + 1) + "/" + year),
                    c.get(Calendar.YEAR), c.get(Calendar.MONTH), c.get(Calendar.DAY_OF_MONTH)).show();
        });
        // 6. Submit Logic
```

```
        btnSubmit.setOnClickListener(v -> {
            if (!cbTerms.isChecked()) {
                Toast.makeText(this, "Please agree to Terms", Toast.LENGTH_SHORT).show();
                return;
            }
            String password = etPass.getText().toString();
            String confirm = etConfirm.getText().toString();
            if (password.isEmpty()) {
                Toast.makeText(this, "Password cannot be empty", Toast.LENGTH_SHORT).show();
            } else if (password.equals(confirm)) {
                Toast.makeText(this, "Registration Successful!", Toast.LENGTH_LONG).show();
            } else {
                Toast.makeText(this, "Passwords do not match", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```



- **Write a program that creates a Service to download a file in the background. While the file is downloading, show a Notification indicating the progress. Once the download is complete, the notification should be updated to show Download Complete.**
-
- **Write a program that uses a Service to perform a task in the background, and once the task is completed, it sends a broadcast. A BroadcastReceiver will listen for the\ completion broadcast and display a Notification when the task is done.**

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical"
  android:padding="20dp"
  android:background="#F8F9FA">

  <ImageView
    android:layout_width="100dp"
    android:layout_height="100dp"
    android:src="@android:drawable/stat_sys_download"
    android:layout_marginBottom="20dp"
    android:contentDescription="Download Icon" />

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Background Service Demo"
    android:textSize="22sp"
    android:textStyle="bold"
    android:textColor="#333"
    android:layout_marginBottom="10dp" />

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click the button below to start a simulated background download with progress notifications."
    android:gravity="center"
    android:textSize="14sp"
    android:layout_marginBottom="40dp"
    android:textColor="#666" />
```

```
    <Button
        android:id="@+id/btnDownload"
        android:layout_width="match_parent"
        android:layout_height="60dp"
        android:text="START DOWNLOAD"
        android:textSize="16sp"
        android:backgroundTint="#2196F3"
        android:textColor="#FFFFFF" />

</LinearLayout>
```

MainActivity’

```
package com.example.downloadapp;

import android.Manifest;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.content.IntentFilter;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import android.app.NotificationManager;

public class MainActivity extends AppCompatActivity {

    private final BroadcastReceiver downloadReceiver = new BroadcastReceiver() {
        @Override
        public void onReceive(Context context, Intent intent) {
            if ("com.example.DOWNLOAD_COMPLETE".equals(intent.getAction())) {
                showFinalNotification(context);
                Toast.makeText(context, "Broadcast Received: Task Done!", Toast.LENGTH_LONG).show();
            }
        }
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // 1. Initialize the Button
    Button btnStart = findViewById(R.id.btnDownload);

    // 2. Request Notification Permission (Crucial for Android 13+)
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS) !=
PackageManager.PERMISSION_GRANTED) {
            requestPermissions(new String[]{Manifest.permission.POST_NOTIFICATIONS}, 101);
        }
    }

    btnStart.setOnClickListener(v -> {
        Intent intent = new Intent(this, DownloadService.class);
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            startForegroundService(intent);
        } else {
            startService(intent);
        }
    });
  }

  private void showFinalNotification(Context context) {
    NotificationManager manager = (NotificationManager) context.getSystemService(NOTIFICATION_SERVICE);
    NotificationCompat.Builder builder = new NotificationCompat.Builder(context, "DownloadChannel")
            .setSmallIcon(android.R.drawable.stat_sys_download_done)
            .setContentTitle("Success!")
            .setContentText("The background task has finished successfully.")
            .setPriority(NotificationCompat.PRIORITY_HIGH)
            .setAutoCancel(true); // Allows notification to disappear when clicked

    manager.notify(2, builder.build());
  }

  @Override
  protected void onStart() {
    super.onStart();
    IntentFilter filter = new IntentFilter("com.example.DOWNLOAD_COMPLETE");
    // Use RECEIVER_NOT_EXPORTED for internal app broadcasts on Android 14
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        registerReceiver(downloadReceiver, filter, Context.RECEIVER_NOT_EXPORTED);
    } else {
```

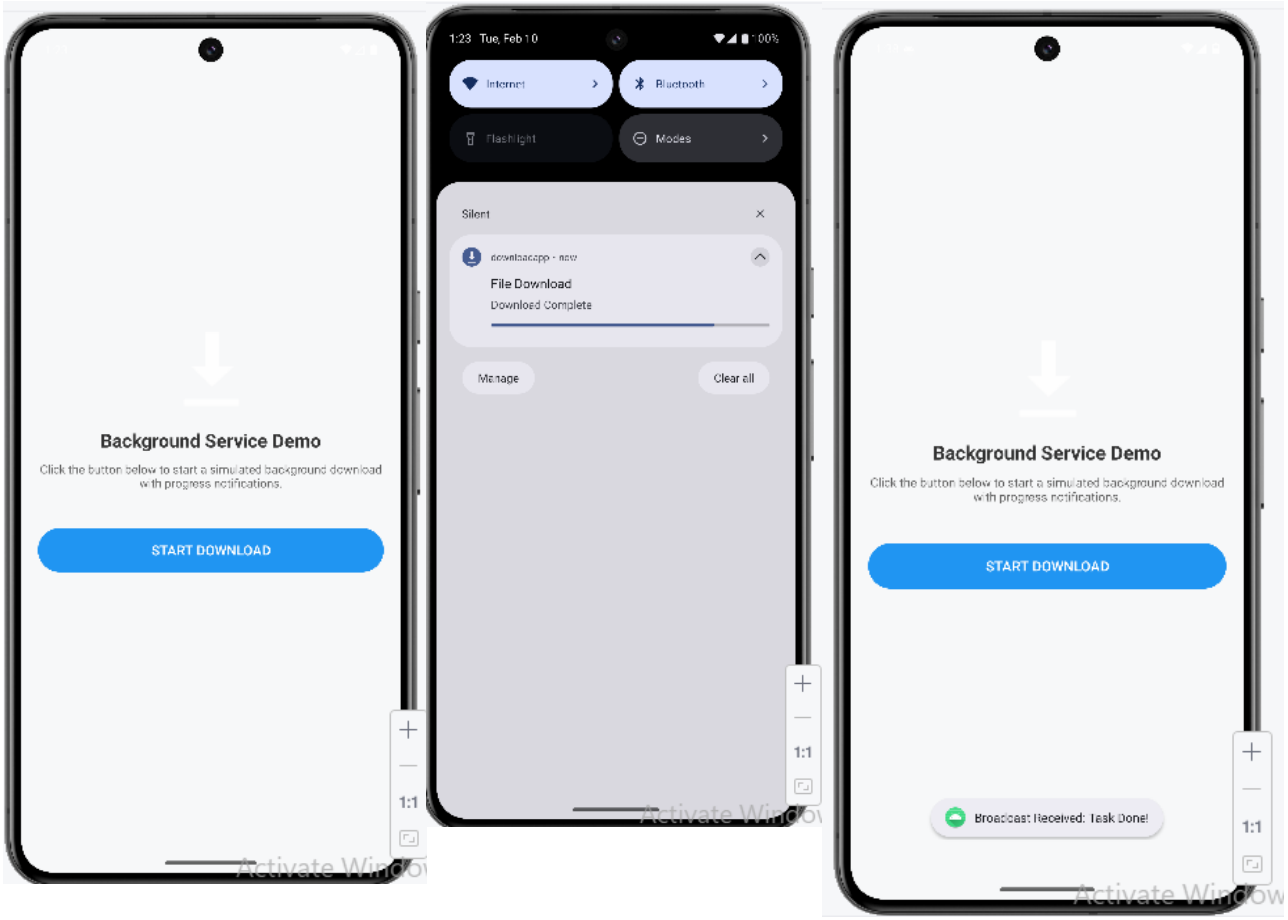
```
        registerReceiver(downloadReceiver, filter);
    }
  }

  @Override
  protected void onStop() {
    super.onStop();
    unregisterReceiver(downloadReceiver);
  }
}
```

DownloadService

```
package com.example.downloadapp;

import android.app.Notification;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.Service;
import android.content.Intent;
import android.os.Build;
import android.os.IBinder;
import androidx.core.app.NotificationCompat;

public class DownloadService extends Service {
  private static final String CHANNEL_ID = "DownloadChannel";
  private NotificationManager notificationManager;
  private NotificationCompat.Builder builder;

  @Override
  public void onCreate() {
    super.onCreate();
    createNotificationChannel();
  }

  @Override
  public int onStartCommand(Intent intent, int flags, int startId) {
    notificationManager = (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

    // Initial Notification
    builder = new NotificationCompat.Builder(this, CHANNEL_ID)
        .setContentTitle("File Download")
        .setContentText("Download in progress")
```

```
            .setSmallIcon(android.R.drawable.stat_sys_download)
            .setOngoing(true)
            .setProgress(100, 0, false);

    startForeground(1, builder.build());

    // Simulate Download in a background thread
    new Thread(() -> {
        for (int i = 0; i <= 100; i += 20) {
            try { Thread.sleep(1000); } catch (InterruptedException e) { e.printStackTrace(); }

            // Update progress
            builder.setProgress(100, i, false);
            notificationManager.notify(1, builder.build());
        }

        // 1. Send the broadcast when task is done
        Intent completionIntent = new Intent("com.example.DOWNLOAD_COMPLETE");
        // For modern Android, we set the package to ensure only our app gets it
        completionIntent.setPackage(getPackageName());
        sendBroadcast(completionIntent);

        // Download Complete
        builder.setContentText("Download Complete")
                .setProgress(0, 0, false)
                .setOngoing(false);
        notificationManager.notify(1, builder.build());


        stopForeground(false); // Keep notification but allow it to be dismissed
        stopSelf(); // Stop the service
    }).start();

    return START_NOT_STICKY;
}

private void createNotificationChannel() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        NotificationChannel channel = new NotificationChannel(
                CHANNEL_ID, "Download Service", NotificationManager.IMPORTANCE_LOW);
        getSystemService(NotificationManager.class).createNotificationChannel(channel);
    }
}
```

```
    @Override
    public IBinder onBind(Intent intent) { return null; }
}
```



IMP:In order to display the notification it will ask permission to 'allow' 'dont allow' to send notification
To display this pop up embed the code bellow in you notification related applications.

Check for Runtime Notification Permission

On Android 13 (API 33) and above, a foreground service cannot show a notification unless you explicitly ask the user for permission. If the permission is denied or not asked, the service might fail to start.

```
import android.content.pm.PackageManager;
import android.os.Build;

if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
```

```
  if (checkSelfPermission(android.Manifest.permission.POST_NOTIFICATIONS) !=
PackageManager.PERMISSION_GRANTED) {
      requestPermissions(new String[]{android.Manifest.permission.POST_NOTIFICATIONS}, 101);
  }
}
```

- **Create a BroadcastReceiver that listens for the Android Intent ACTION_BATTERY_CHANGED broadcast and shows a Toast displaying the current battery percentage.**

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Listening for Battery Changes..."
      android:textSize="18sp"
      android:textStyle="italic" />
</LinearLayout>
```

MainACtivity

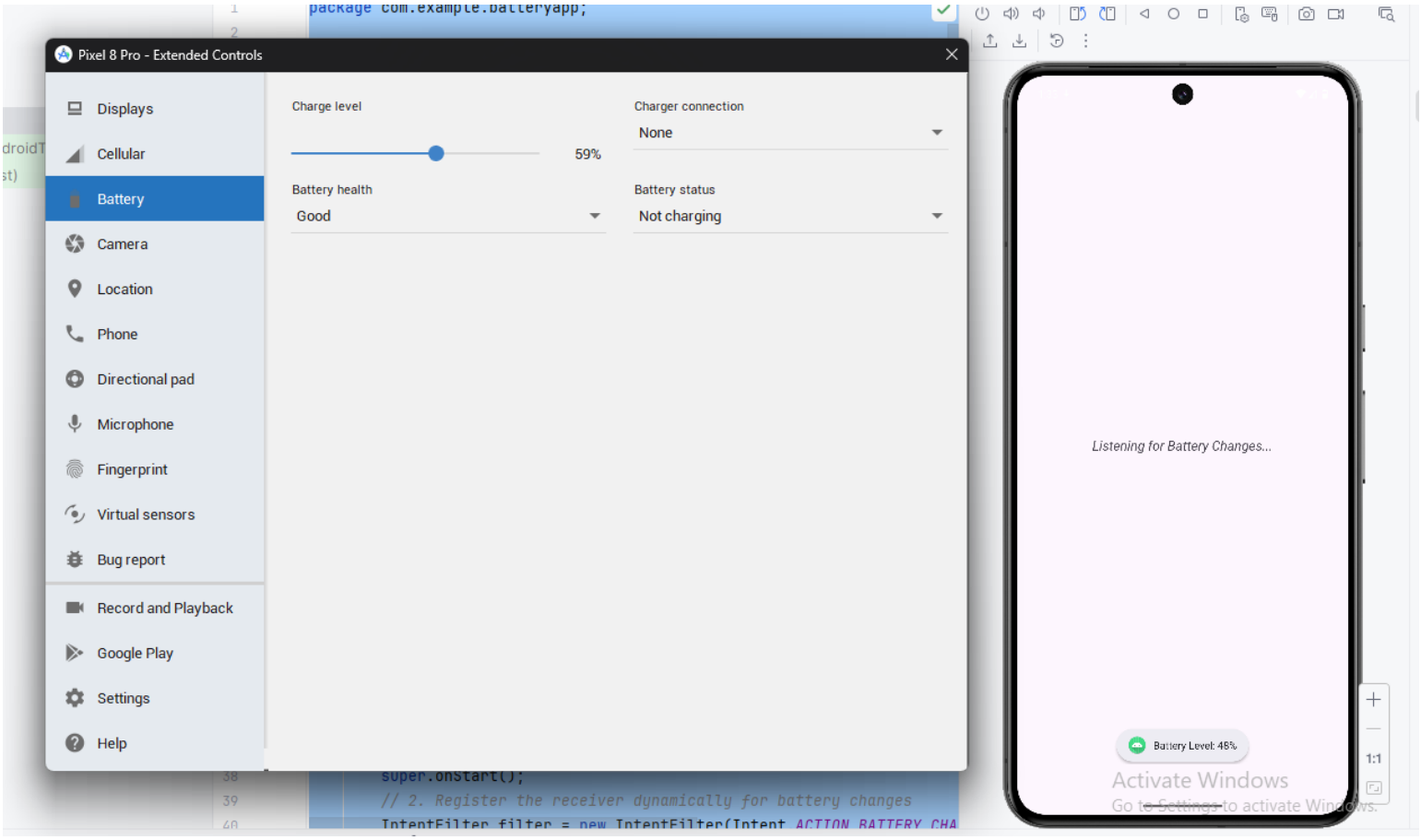
```
package com.example.batteryapp;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.content.IntentFilter;
import android.os.BatteryManager;
import android.os.Bundle;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  // 1. Define the BroadcastReceiver
  private final BroadcastReceiver batteryReceiver = new BroadcastReceiver() {
      @Override
```

```
        public void onReceive(Context context, Intent intent) {
            // Get the current battery level and scale
            int level = intent.getIntExtra(BatteryManager.EXTRA_LEVEL, -1);
            int scale = intent.getIntExtra(BatteryManager.EXTRA_SCALE, -1);

            // Calculate percentage
            float batteryPct = (level / (float) scale) * 100;

            // Show the Toast
            Toast.makeText(context, "Battery Level: " + (int) batteryPct + "%", Toast.LENGTH_SHORT).show();
        }
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    protected void onStart() {
        super.onStart();
        // 2. Register the receiver dynamically for battery changes
        IntentFilter filter = new IntentFilter(Intent.ACTION_BATTERY_CHANGED);
        registerReceiver(batteryReceiver, filter);
    }

    @Override
    protected void onStop() {
        super.onStop();
        // 3. Unregister to prevent memory leaks when the app is not visible
        unregisterReceiver(batteryReceiver);
    }
}
```

IMP: **The Emulator Method (PC)**

If you are using the Android Studio Emulator:

1. Click the **three dots (...)** at the bottom of the emulator sidebar (Extended Controls).
2. Go to the **Battery** tab.
3. Manually move the **Charge level** slider or change the **Battery status** (e.g., from Charging to Discharging).
4. Each time you move that slider, your app's Toast should trigger immediately.


package com.example.batteryapp;
Pixel 8 Pro - Extended Controls
Displays
Cellular
Battery
Camera
Location
Phone
Directional pad
Microphone
Fingerprint
Virtual sensors
Bug report
Record and Playback
Google Play
Settings
Help
Charge level
59%
Charger connection
None
Battery health
Good
Battery status
Not charging
Listening for Battery Changes...
Battery Level: 48%
Activate Windows
Go to Settings to activate Windows.
super.onStart();
// 2. Register the receiver dynamically for battery changes